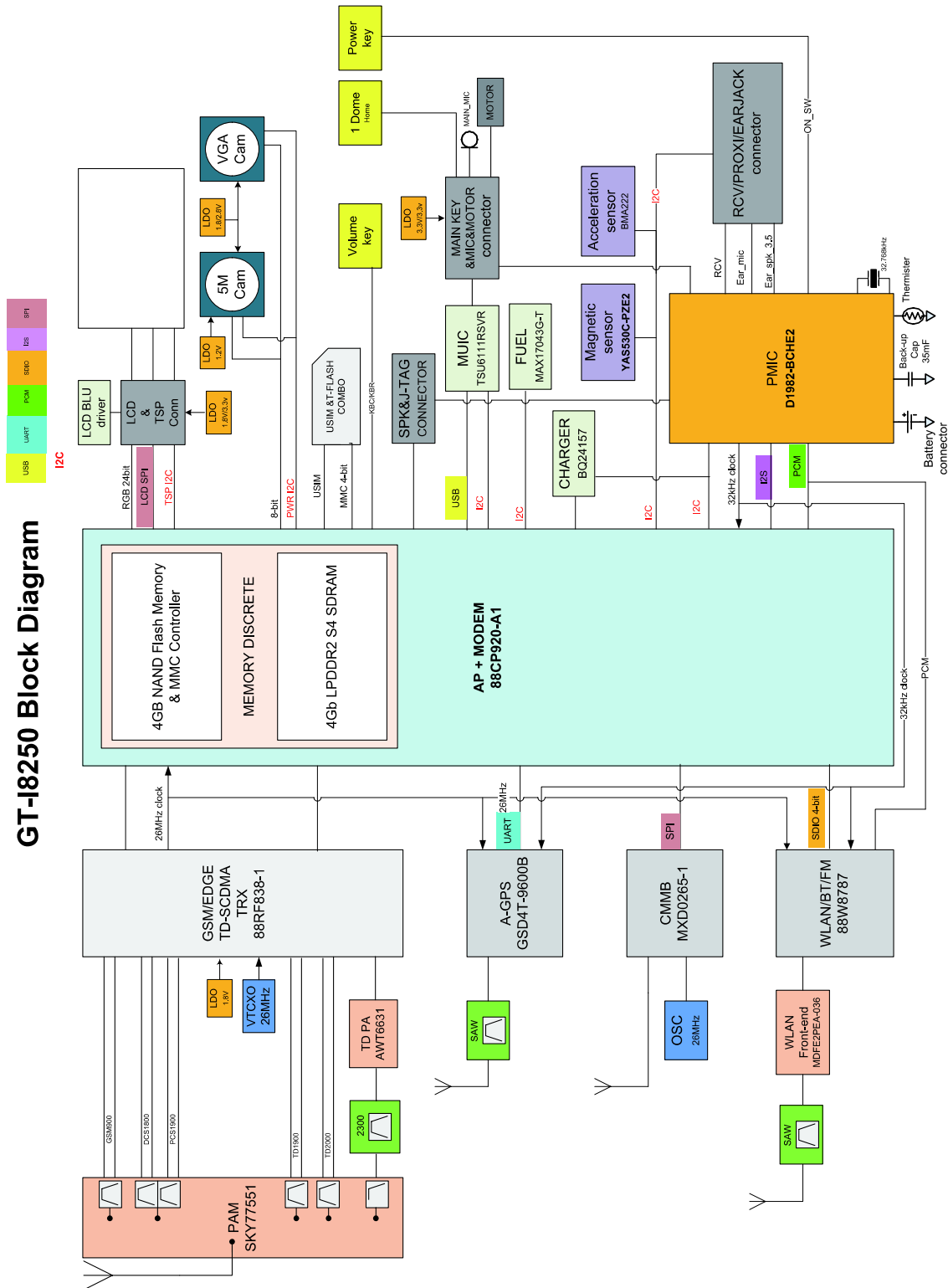


8. Level 3 Repair

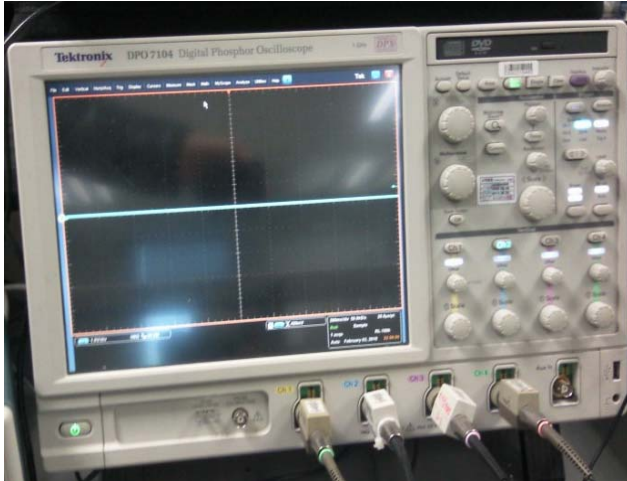
8-1. Block Diagram





8-3. Flow Chart of Troubleshooting

Equipments



↑ Oscilloscope



↑ Digital Multimeter



↑ Power Supply

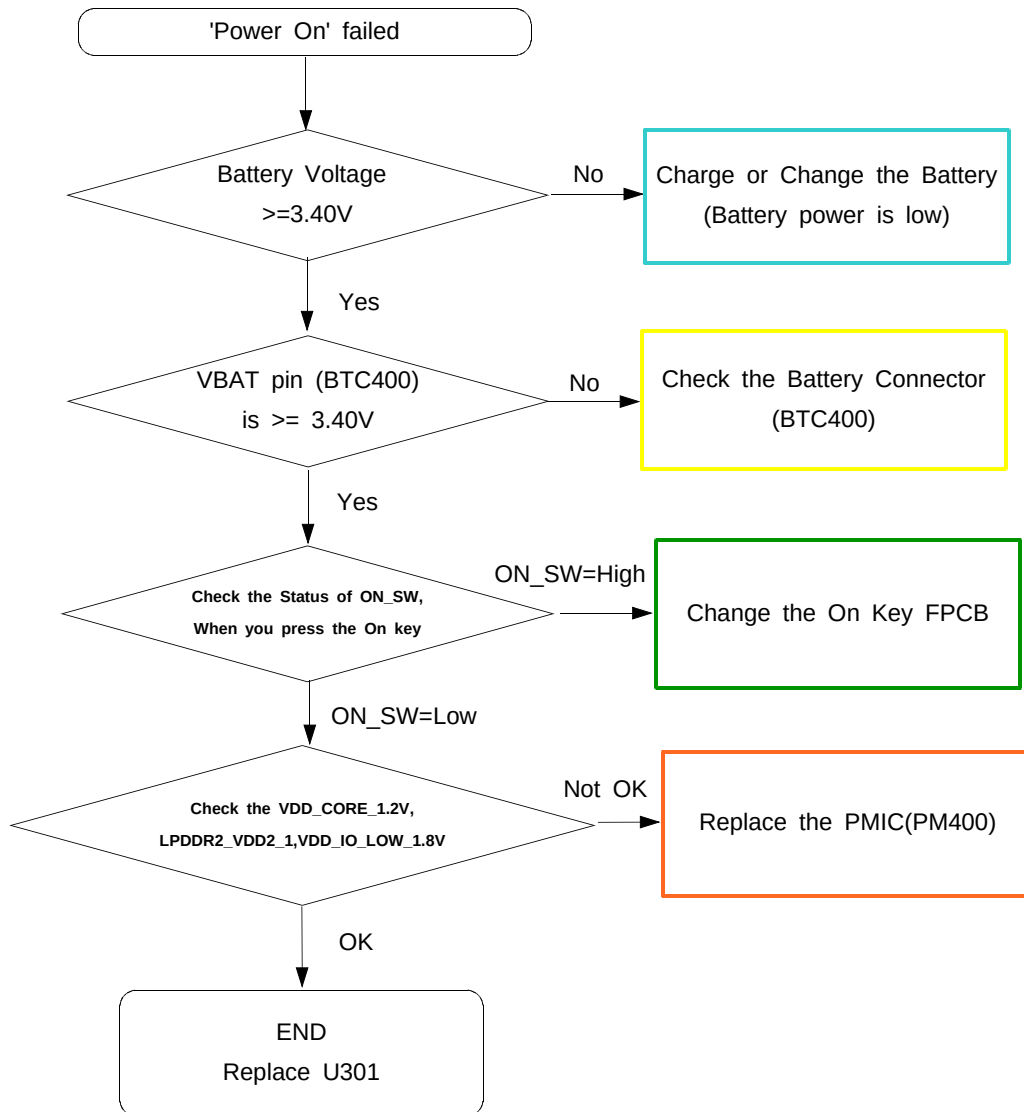


↑ + driver, Tweezer

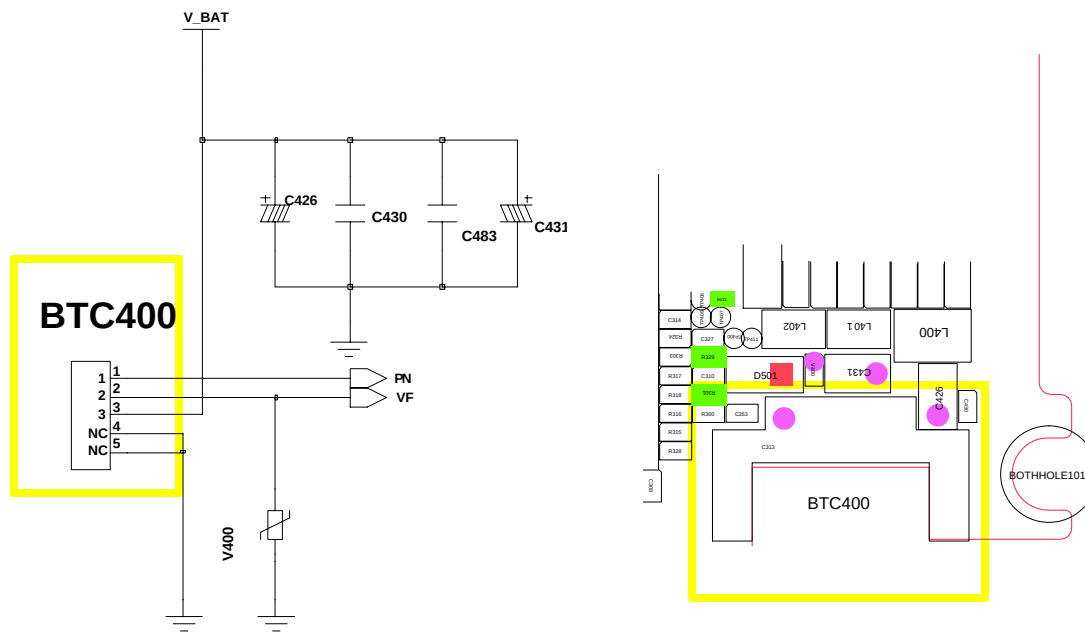
- index

	Part	Duty List	담당자
8-3-1	Power On		조영식
	- PHONE Part		조영식
	- PDA Part		조영식
8-3-2	MIC		홍성우
8-3-3	RECEIVER		홍성우
8-3-4	SPEAKER		홍성우
8-3-5	CMMB		홍일표
8-3-6	GPS		손용호
8-3-7	Bluetooth		고용림
8-3-8	WiFi		고용림
8-3-9	KEY(volume key/ power key / touch key)		안재형
8-3-10	T-Flash		안재형
8-3-11	Camera		조영식
8-3-12	LCD Working(LCD,TSP)		안재형
8-3-13	Sensor		조정민
	Accelerometer(3-Axis)		조정민
	Compass		조정민
	Proximity-Ambient Light		조정민
	RF part		
8-3-14	GSM900 Receiver		손용암
8-3-15	GSM900 Transmitter		손용암
8-3-16	GSM1800 Receiver		손용암
8-3-17	GSM1800 Transmitter		손용암
8-3-18	GSM1900 Receiver		손용암
8-3-19	GSM1900 Transmitter		손용암
8-3-20	TDSCDMA B1,B2 Receiver		손용암
8-3-21	TDSCDMA B1,B2 Transmitter		손용암

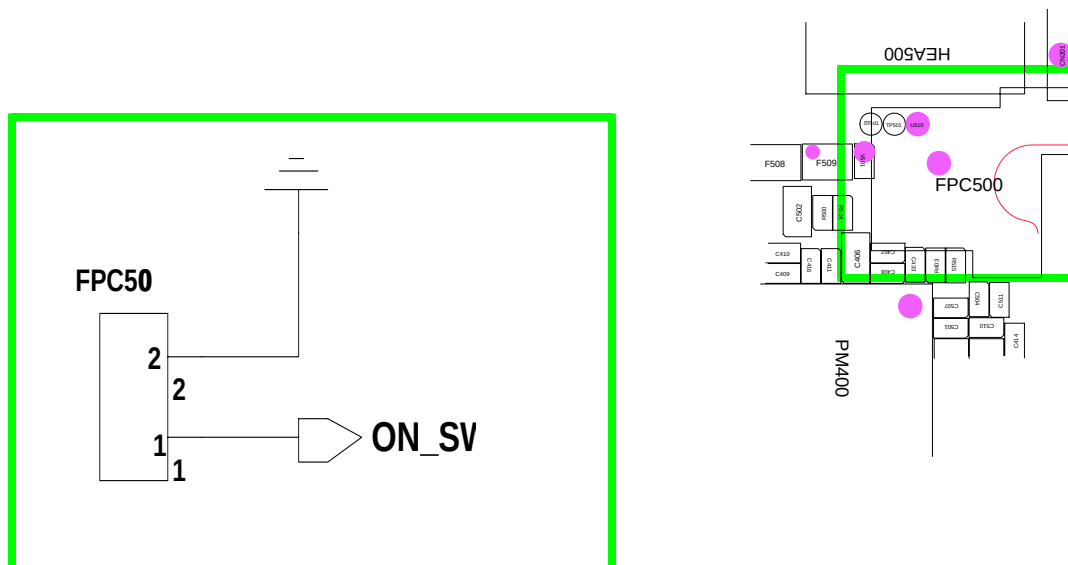
8-3-1. Power On I8250

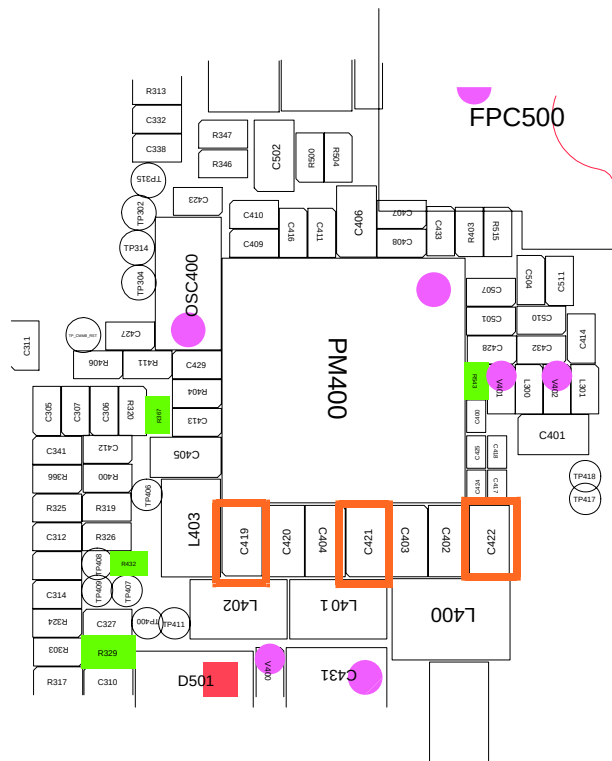


- BTC400

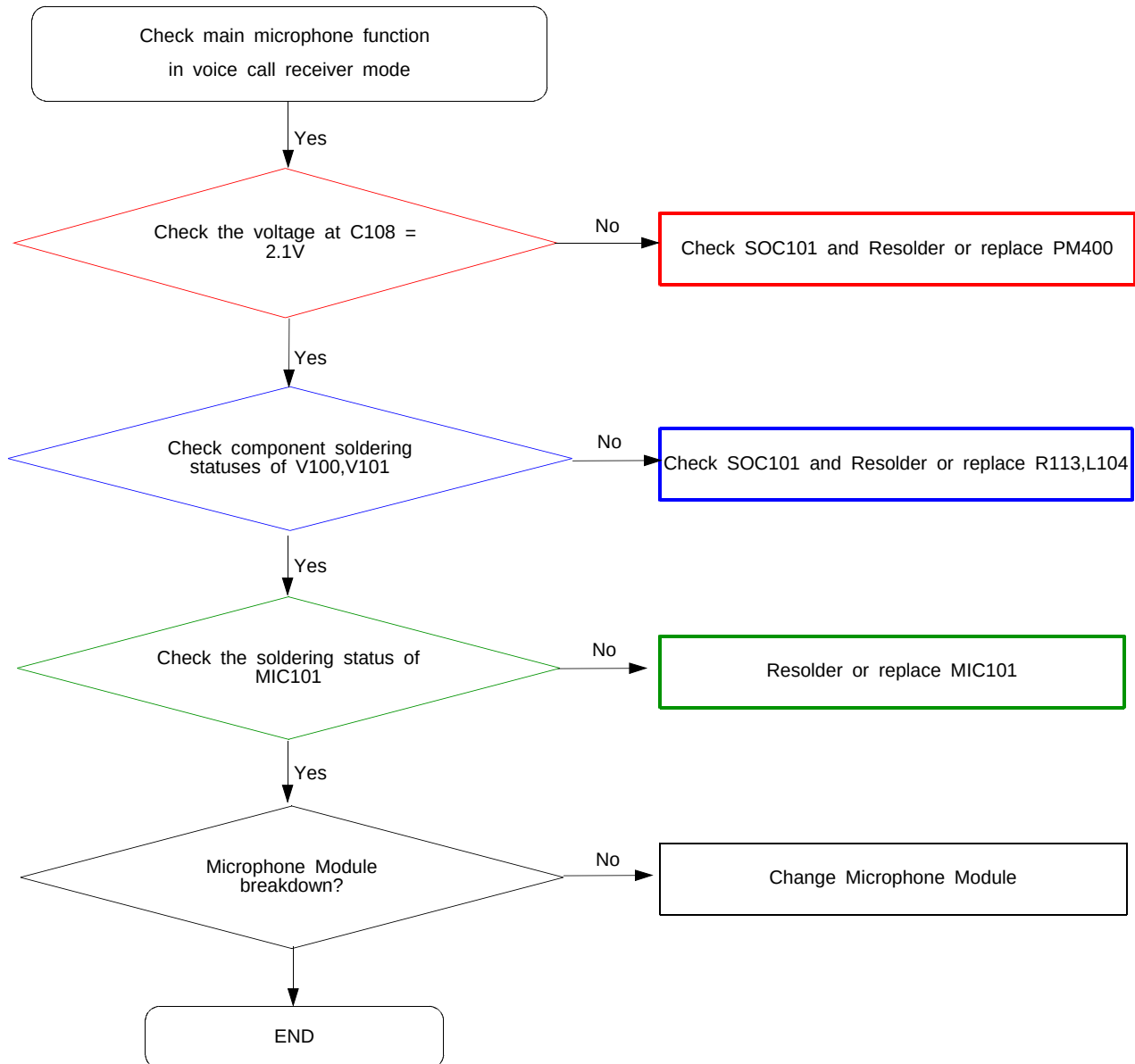


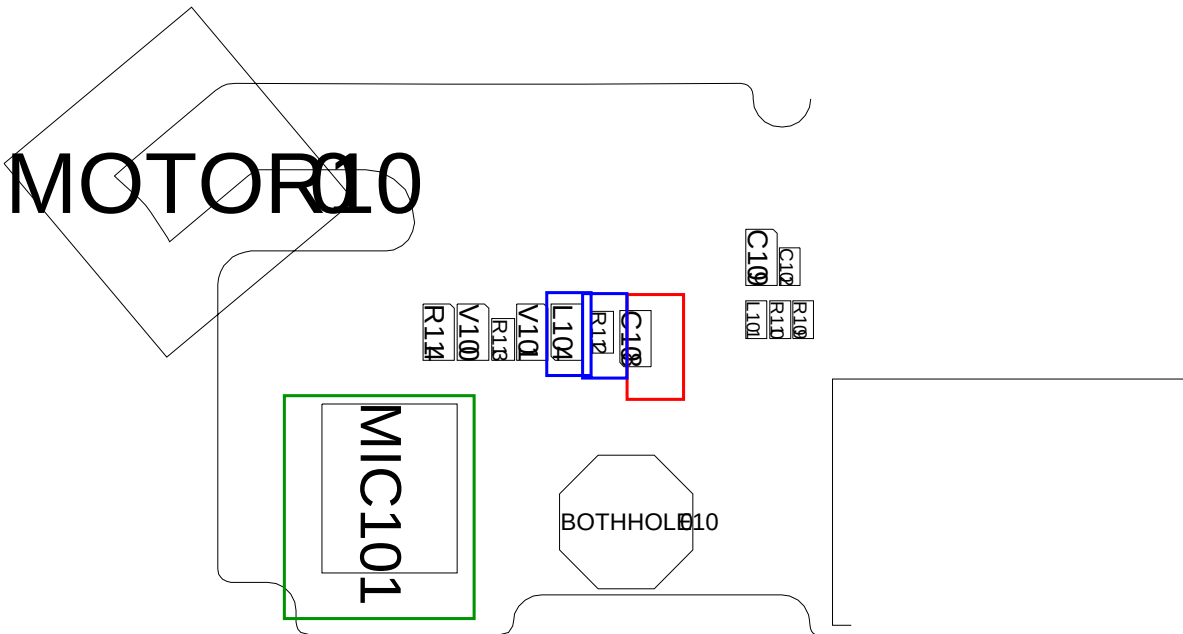
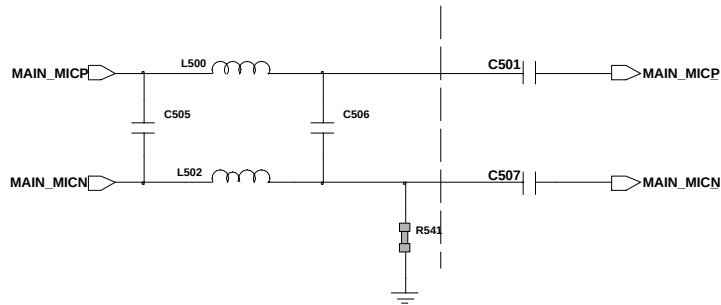
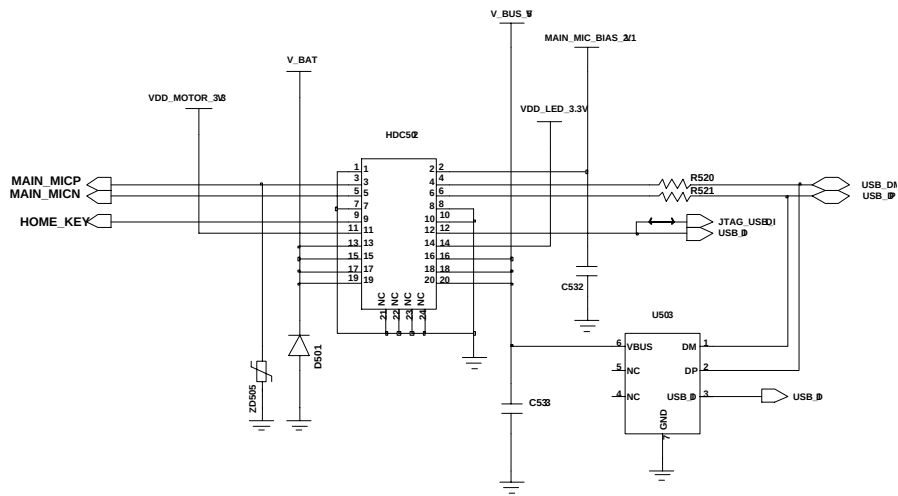
- ON_SW



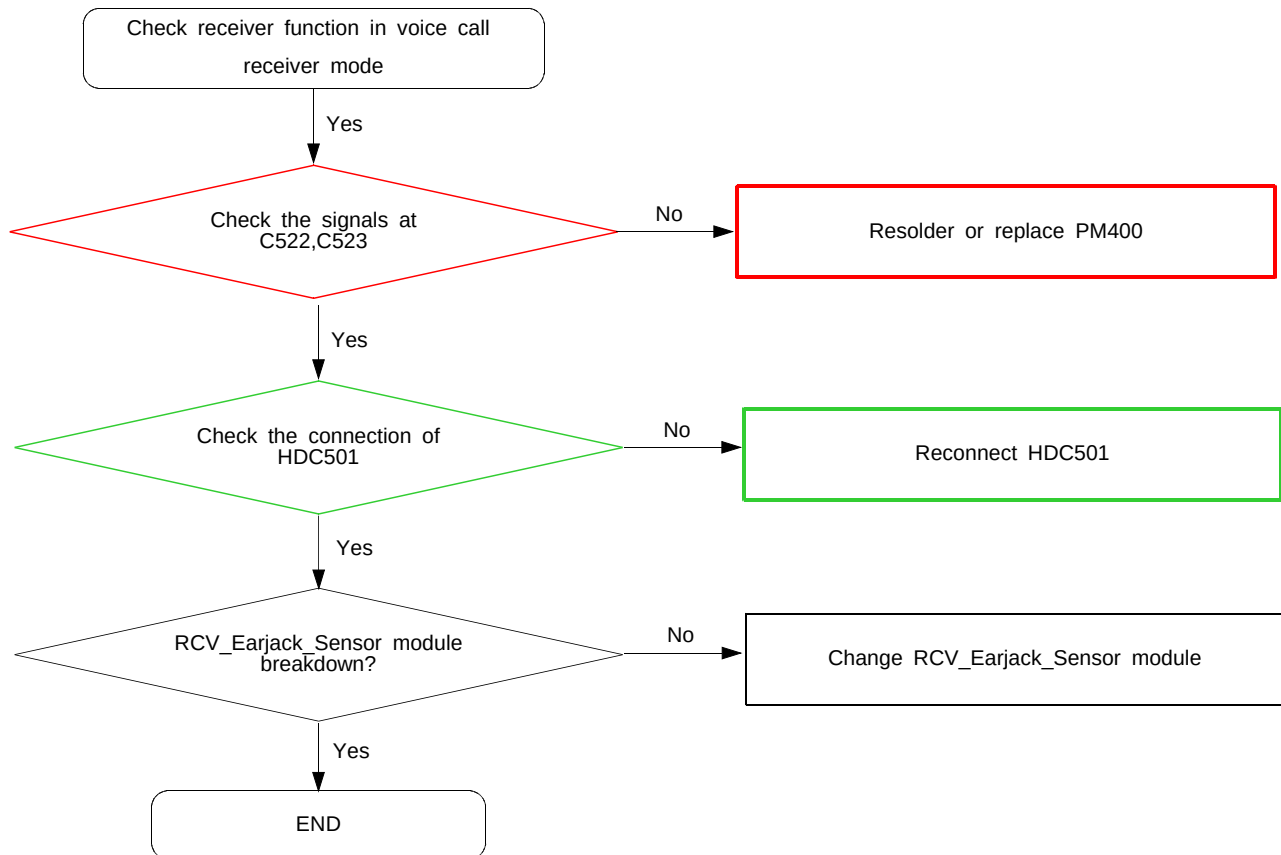


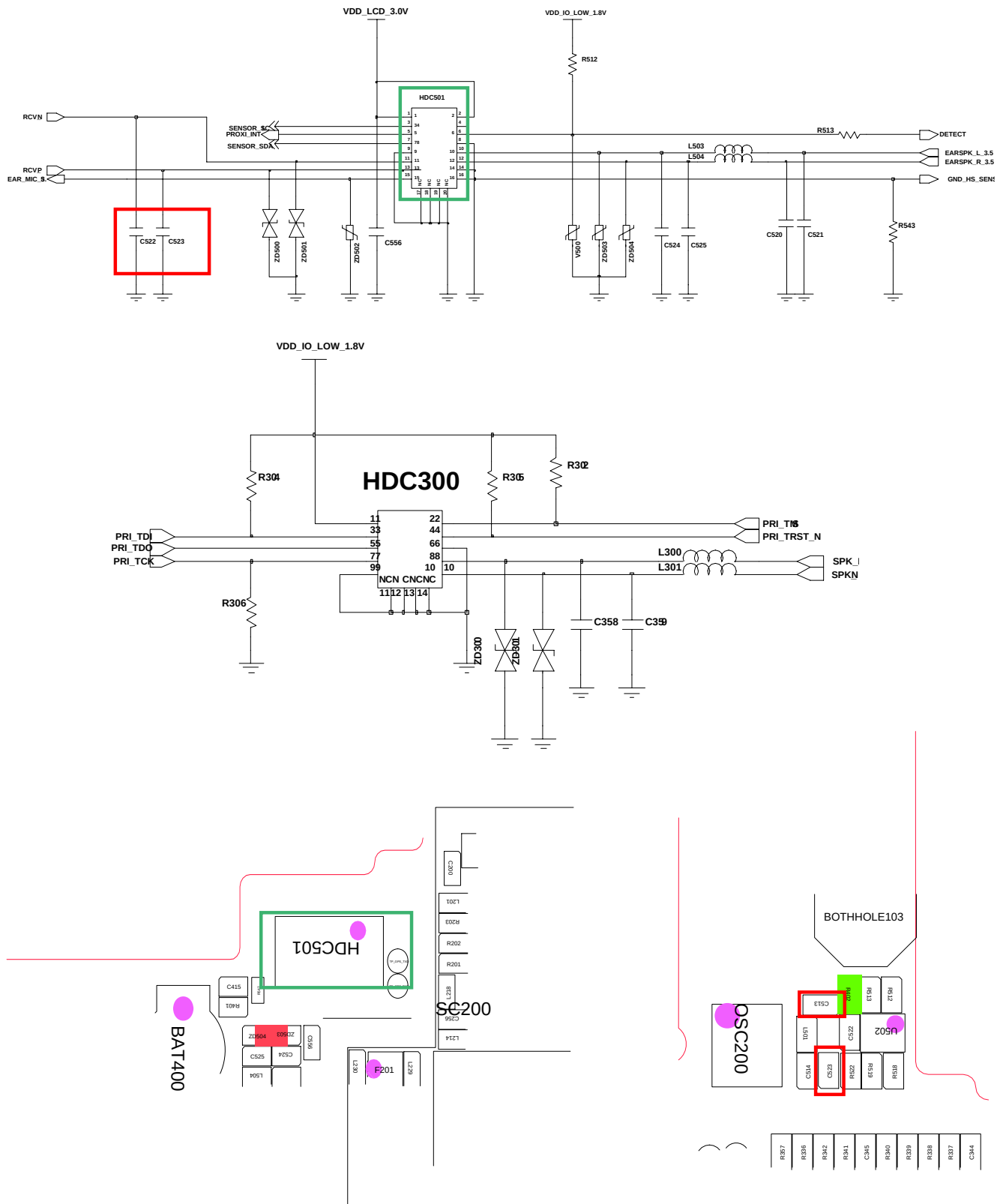
8-3-2. Microphone Part I8250



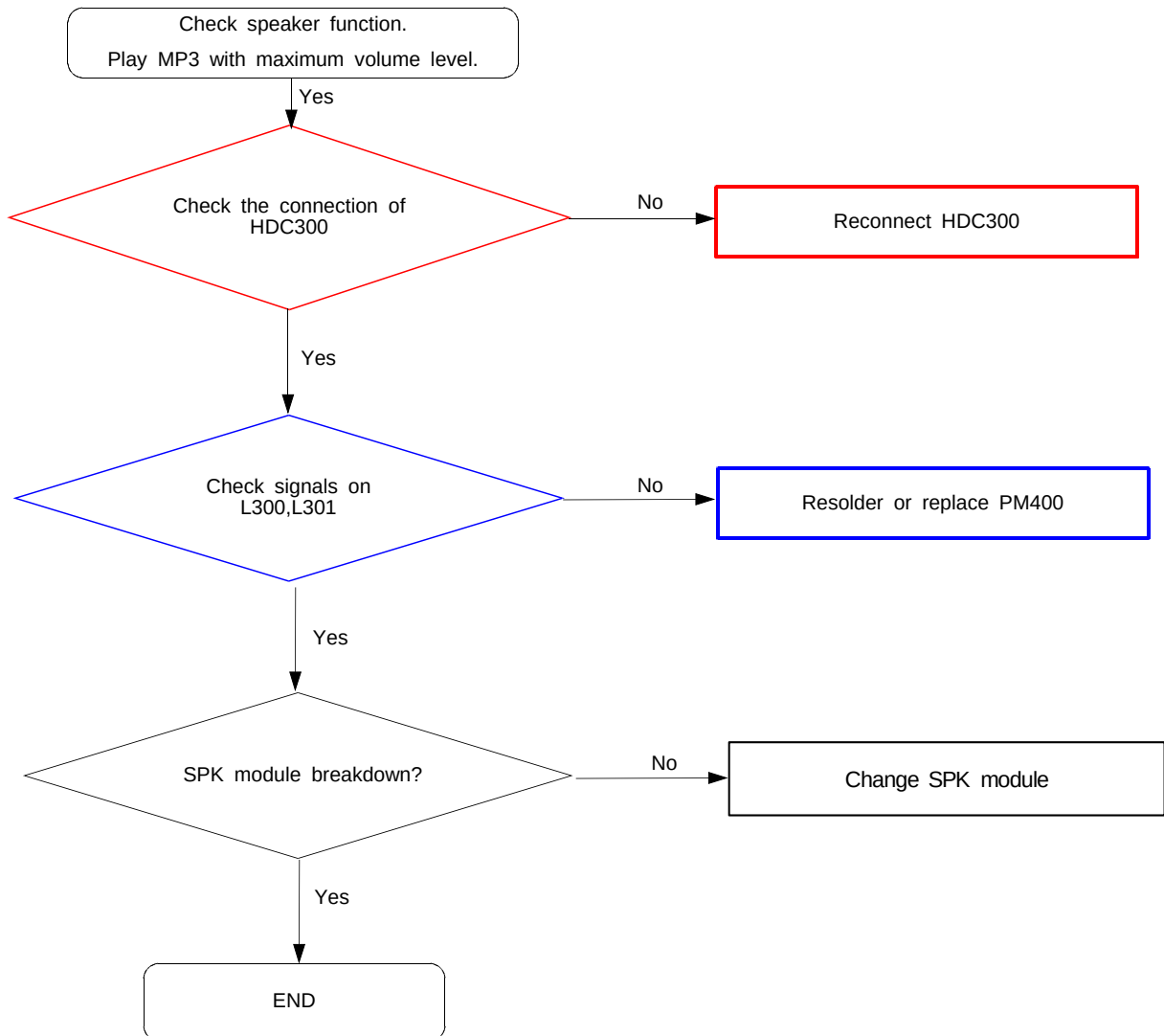


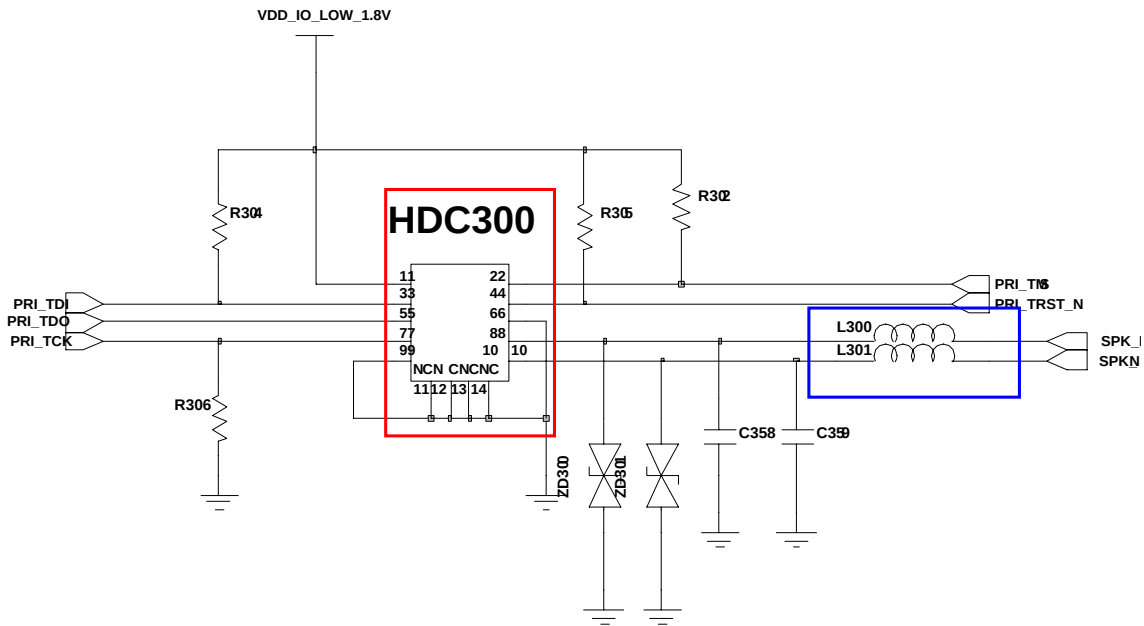
8-3-3. Receiver Part I8250



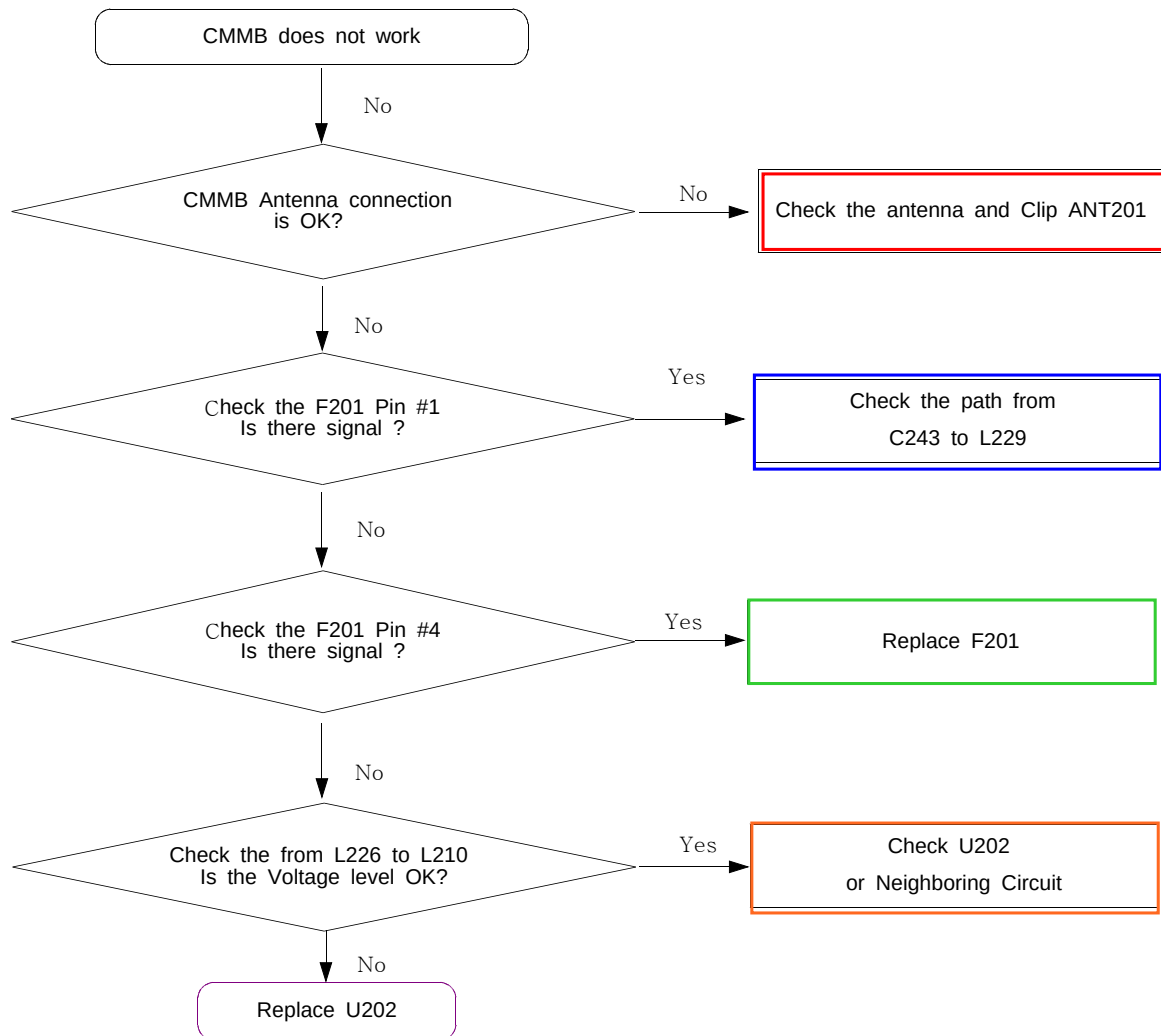


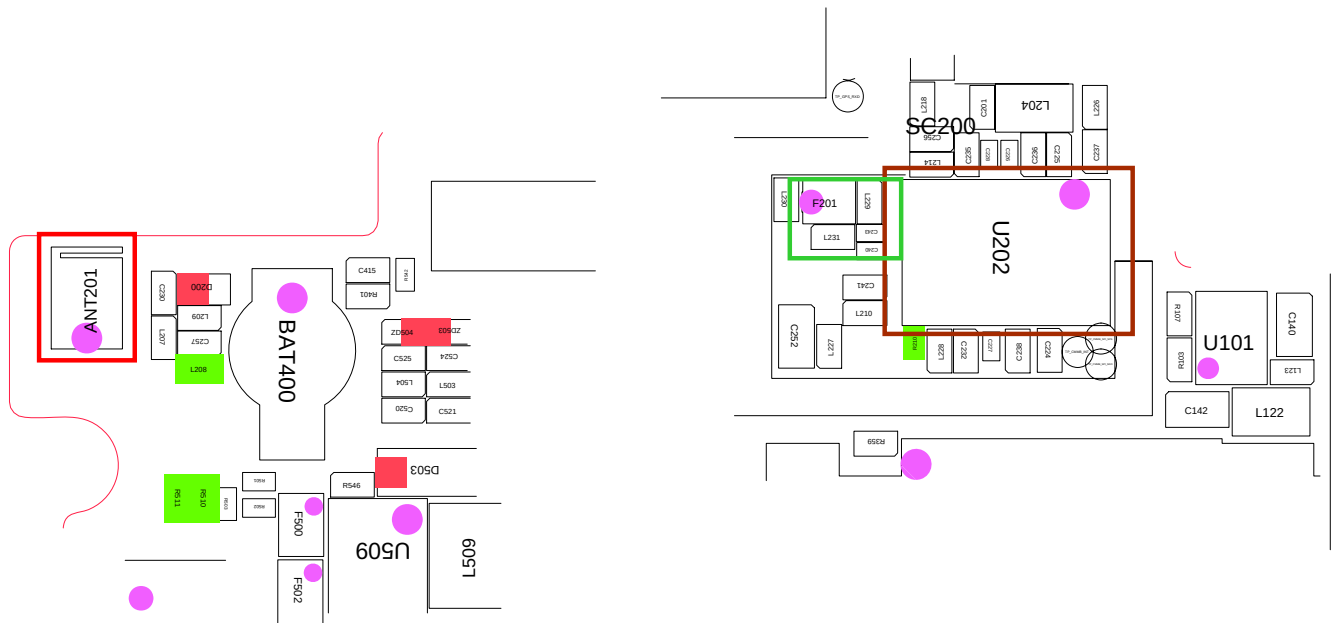
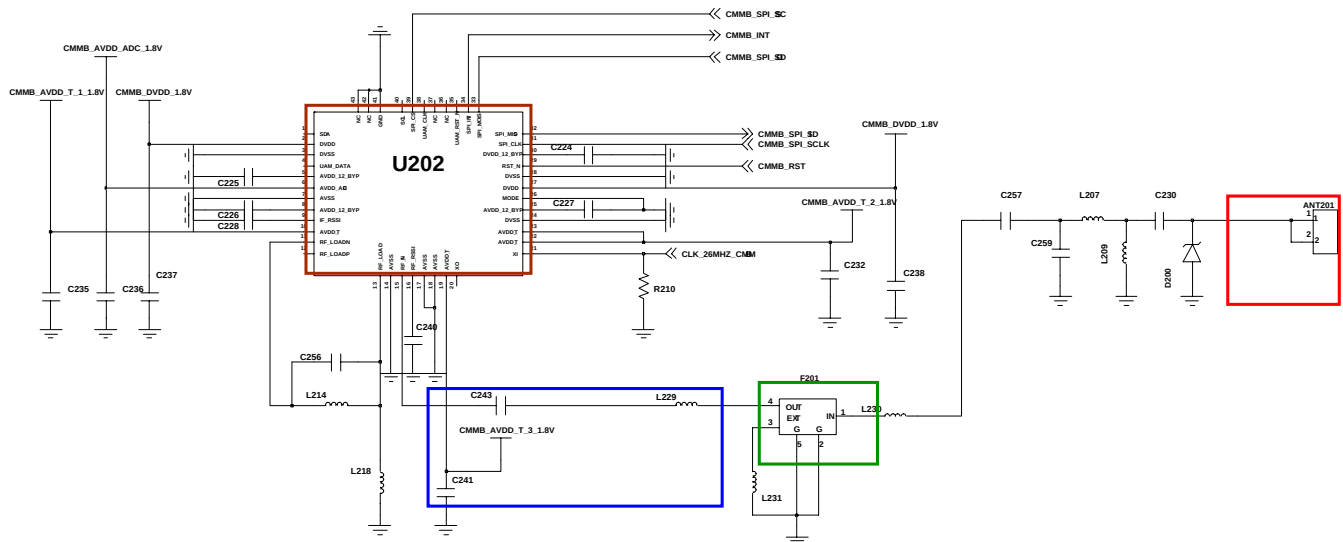
8-3-4. Speaker Part GT-I8250



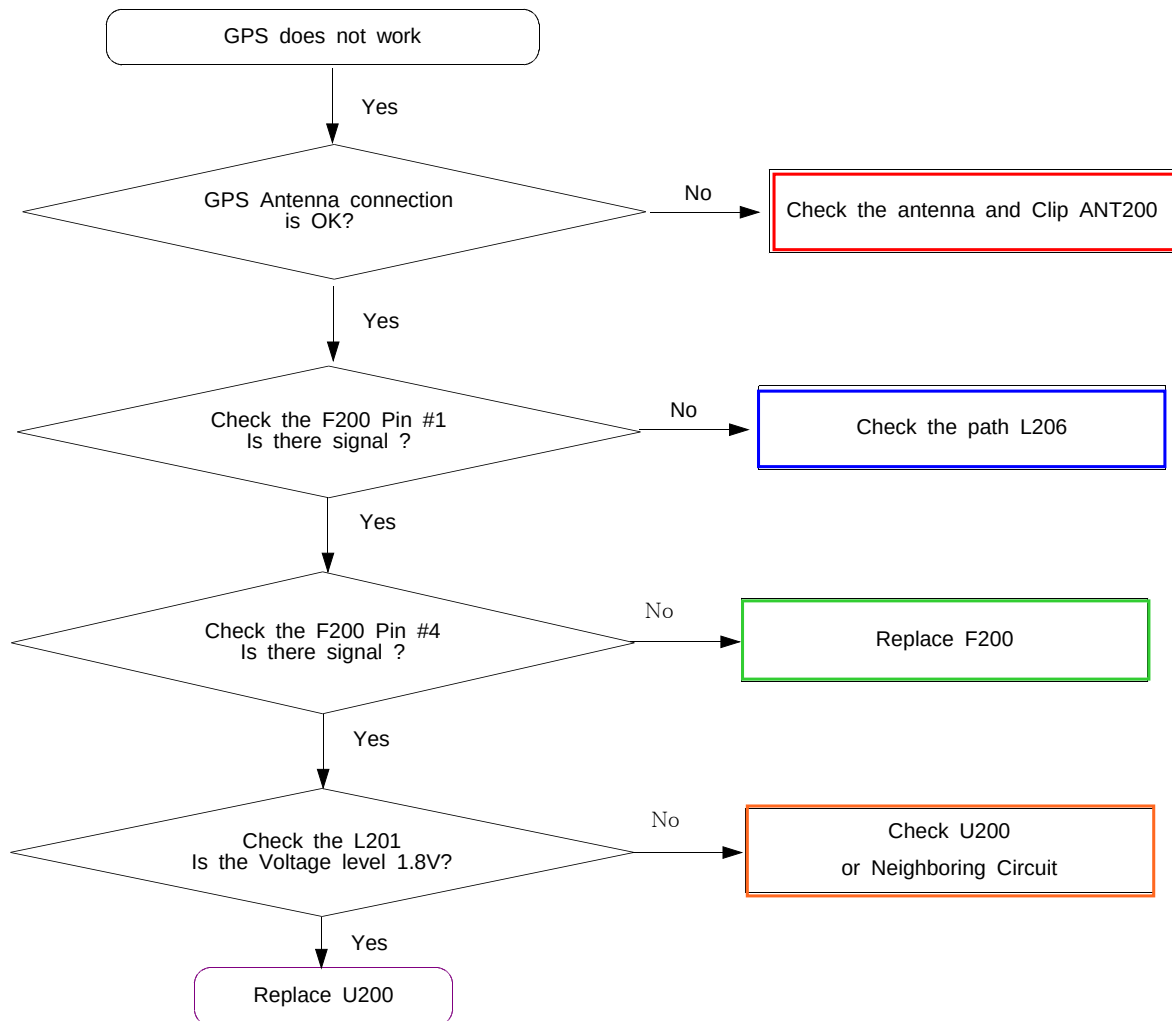


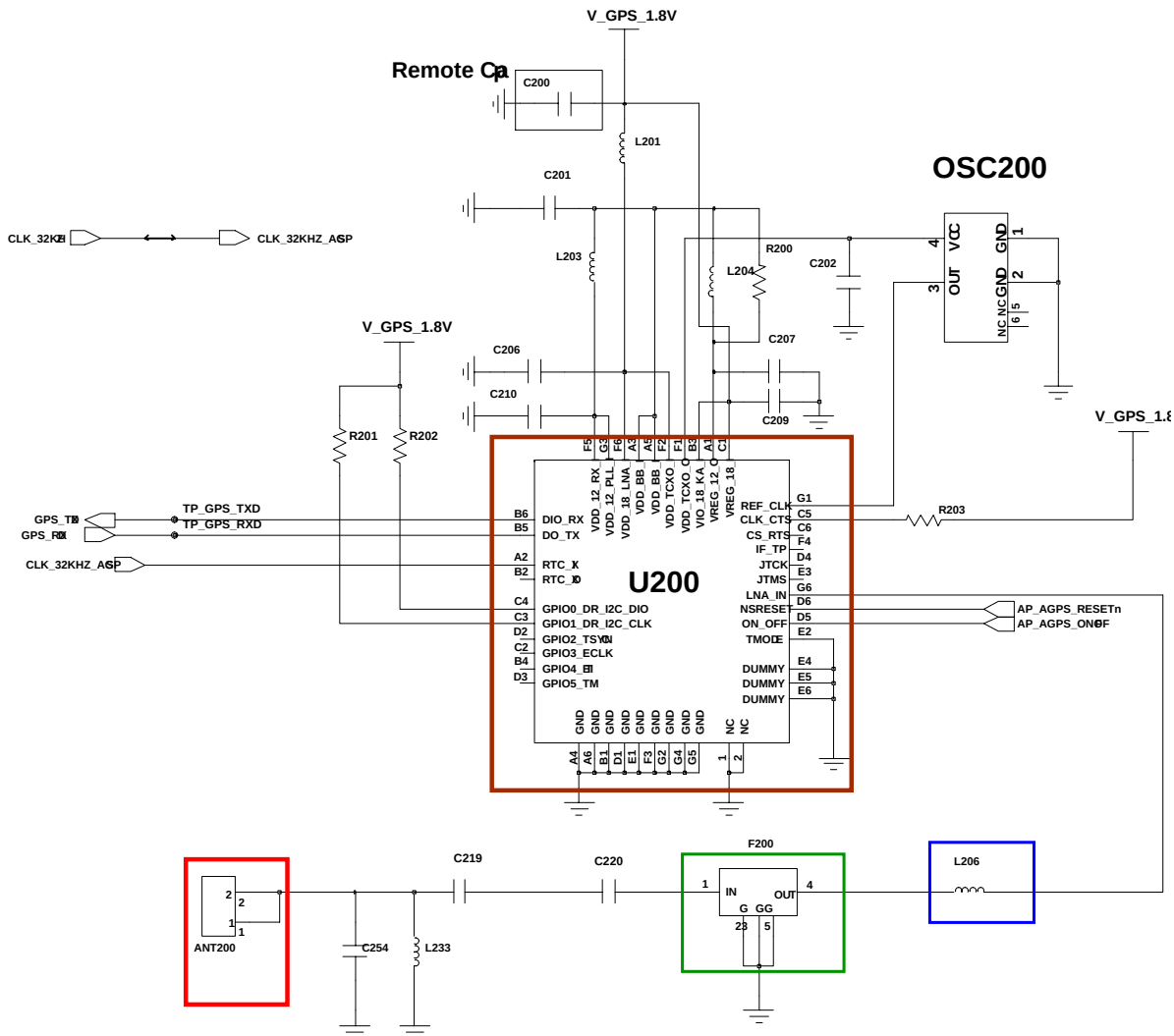
8-3-5 CMMB I8250

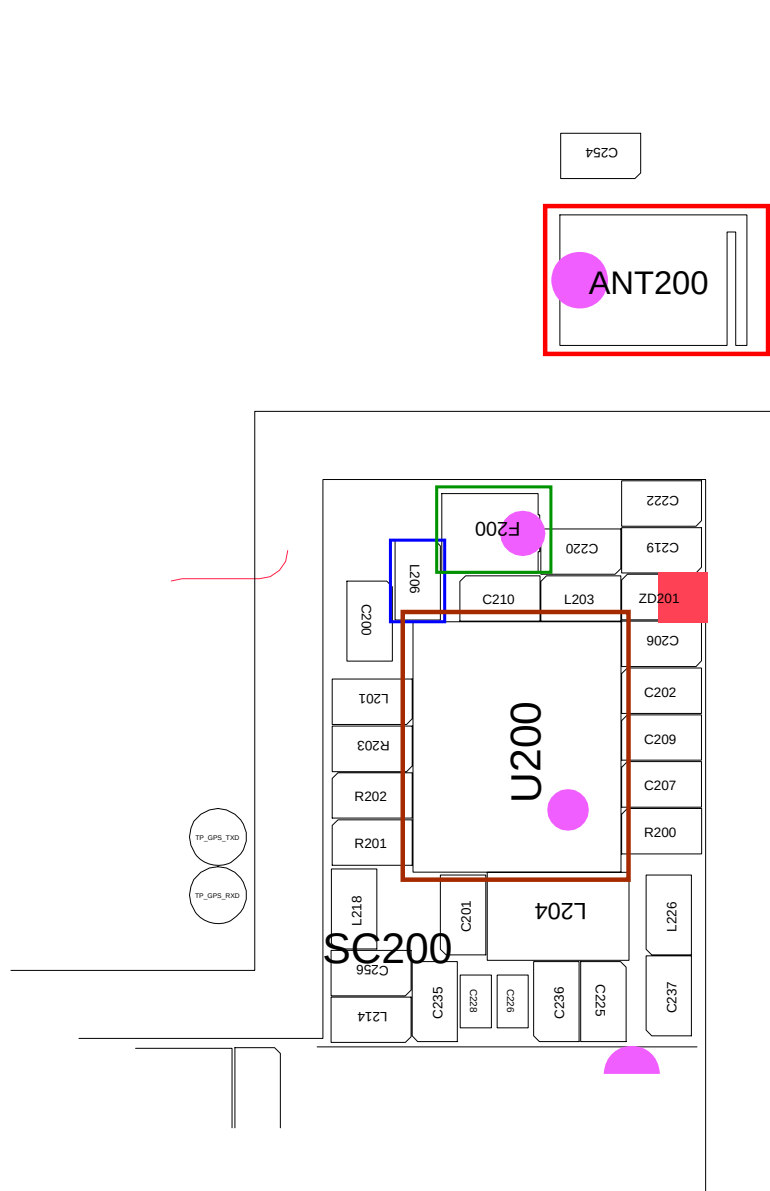




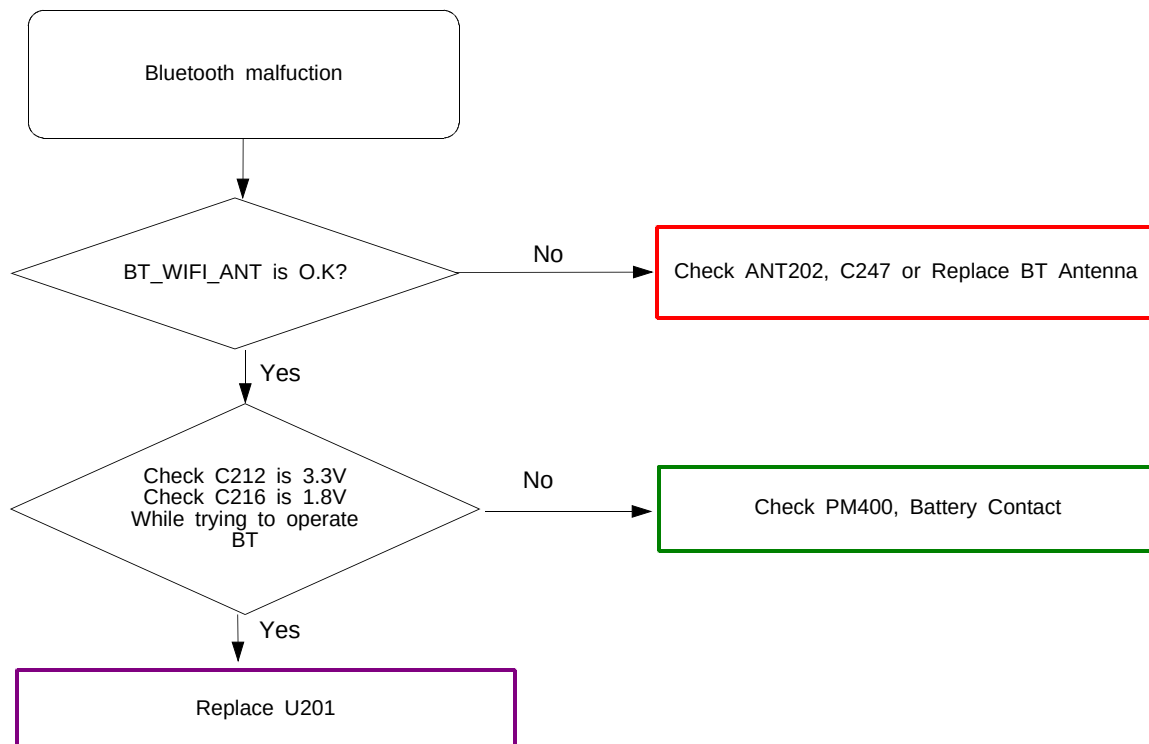
8-3-6 GPS I8250

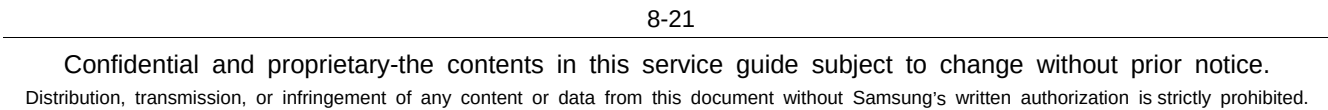


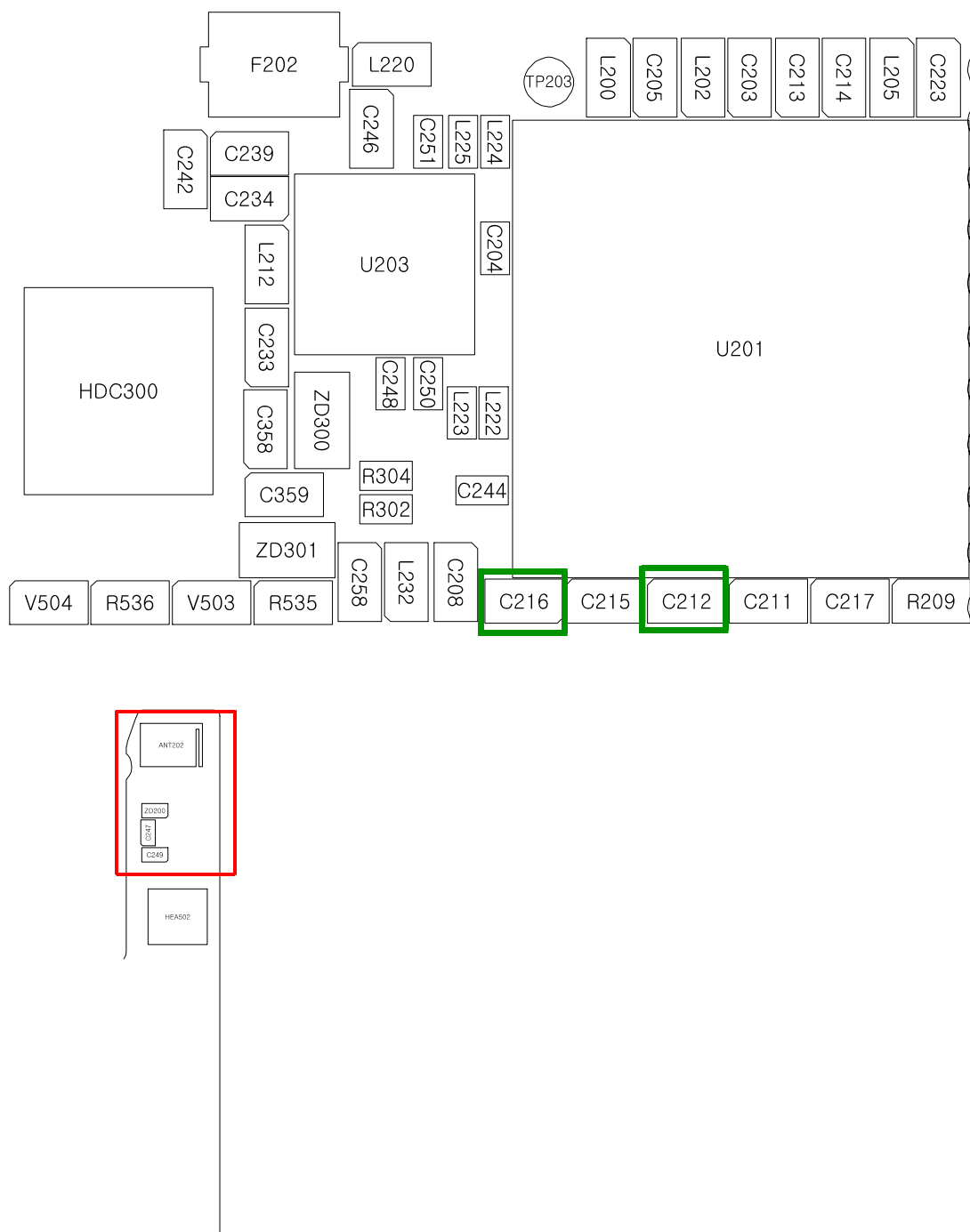




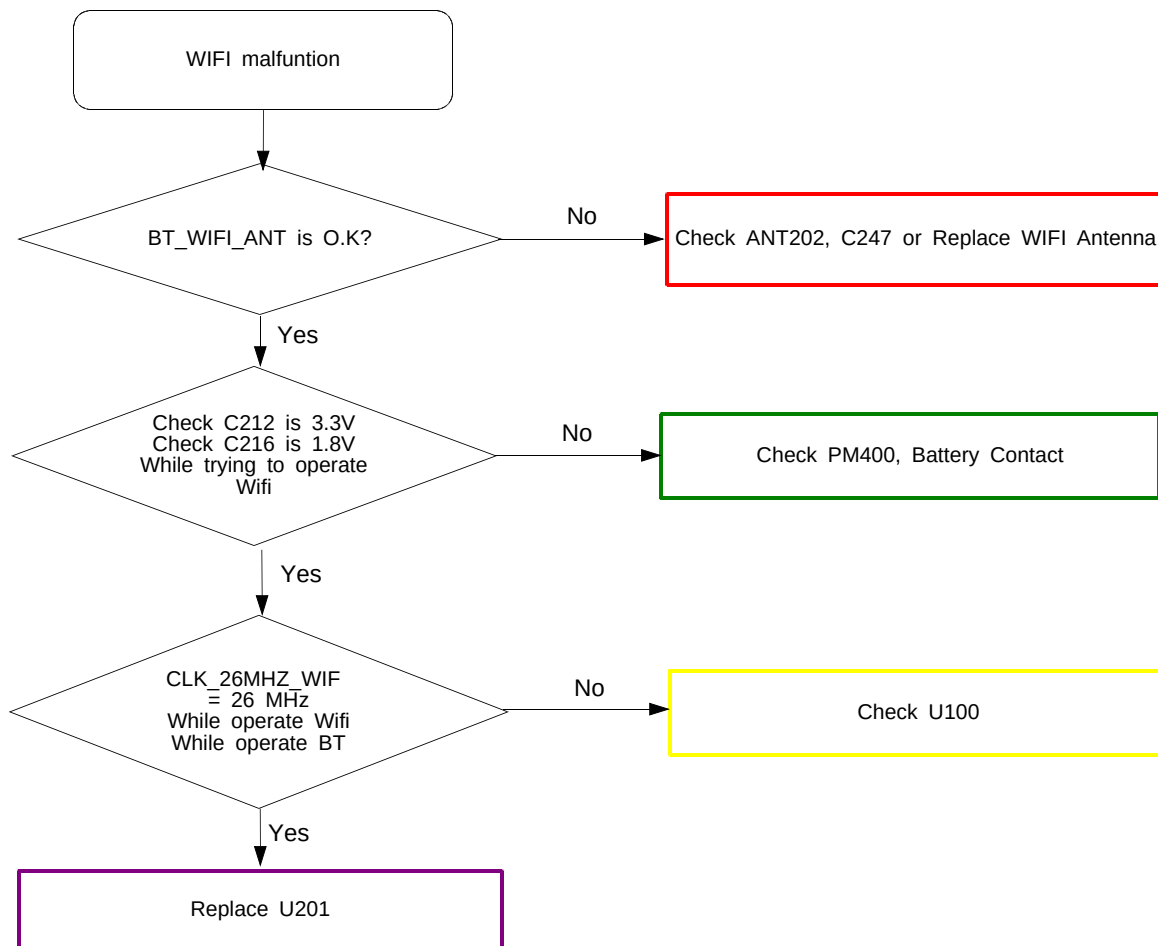
8-3-7. Bluetooth I8250

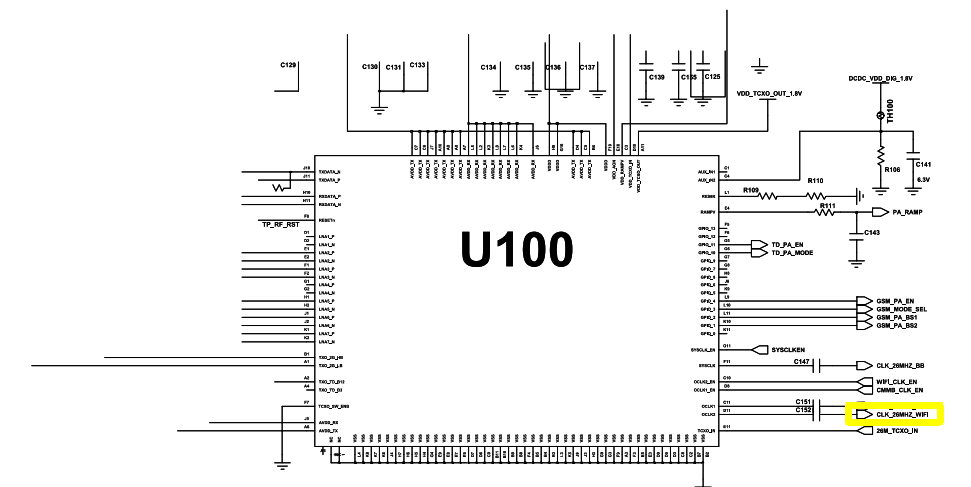


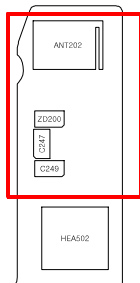
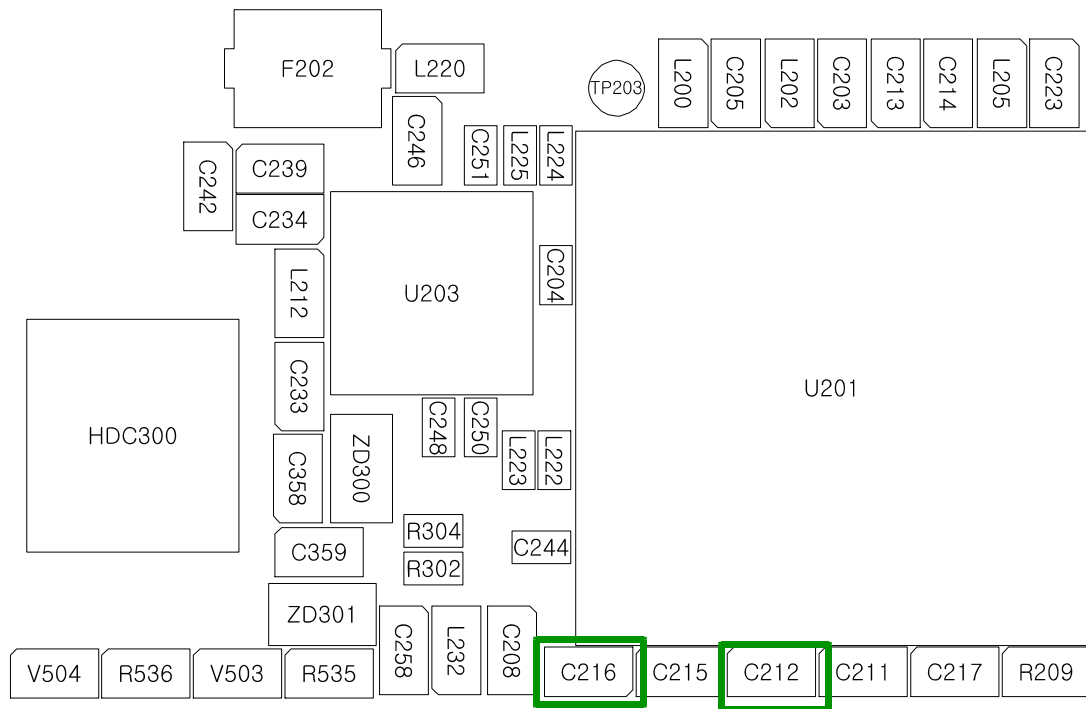




8-3-8. WIFI I8250

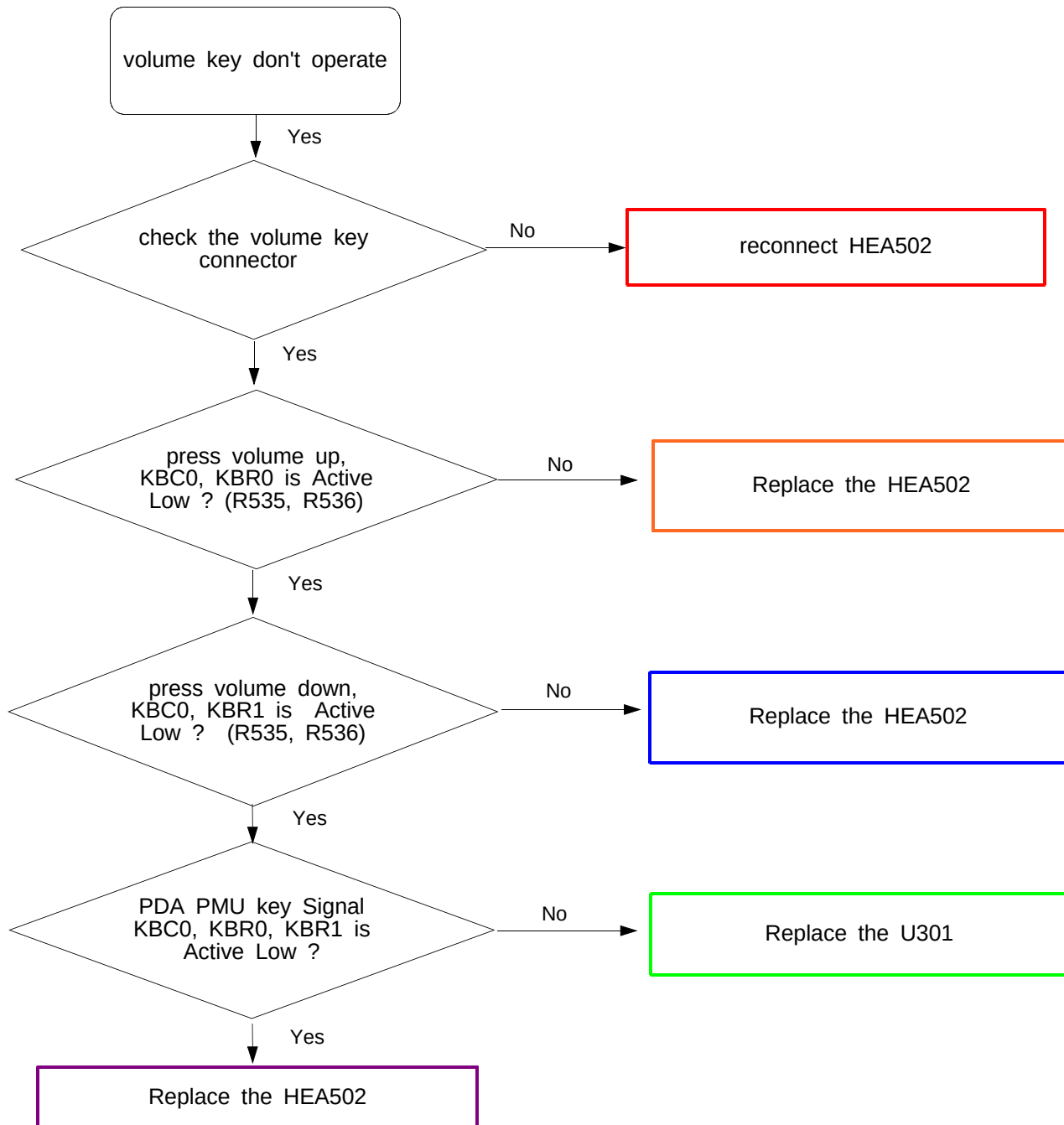


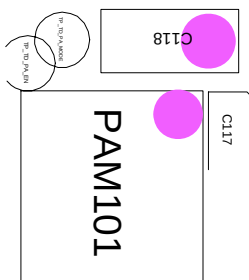
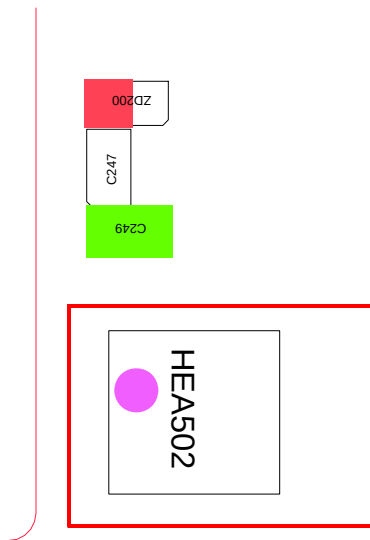
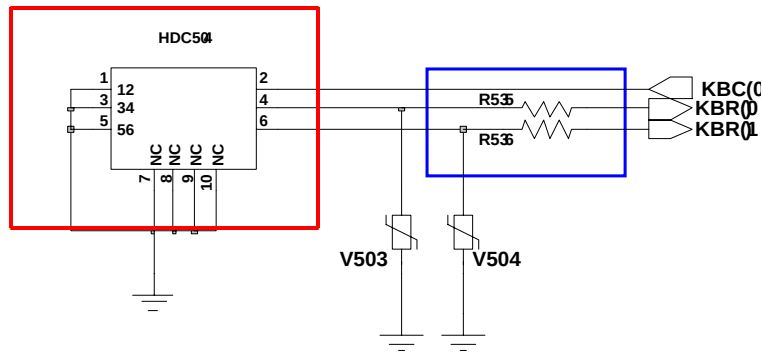




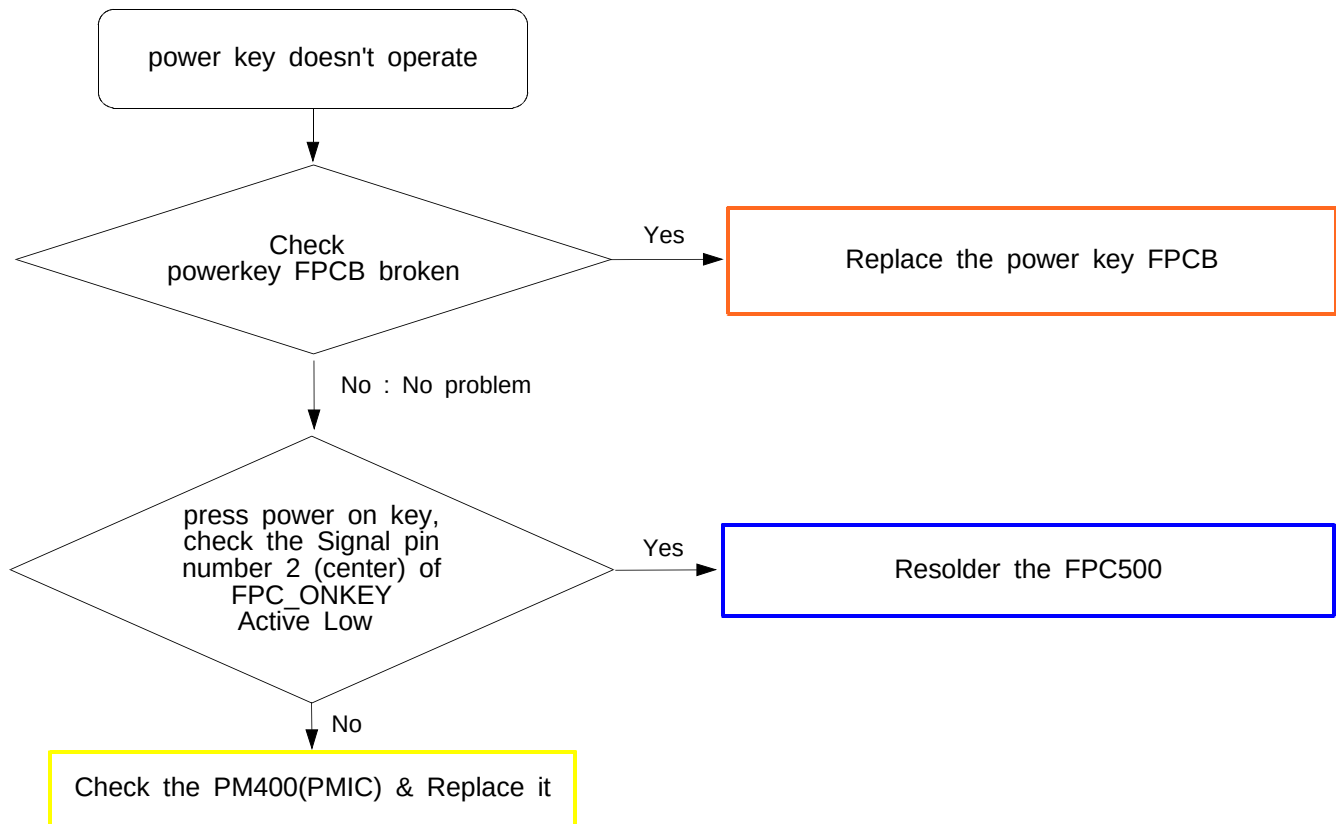
8-3-9. KEY Function (volume key/ power key/ Main Key) GT-I8250

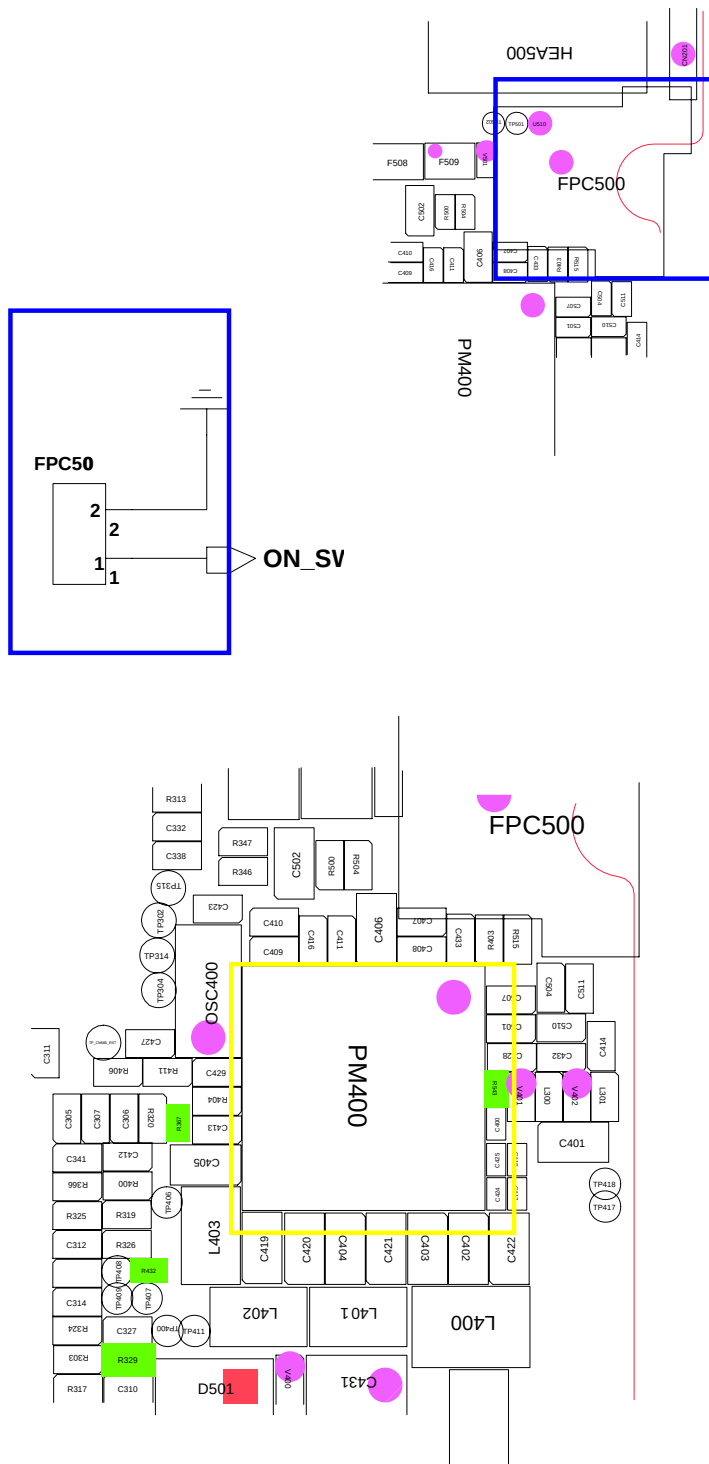
- volume key



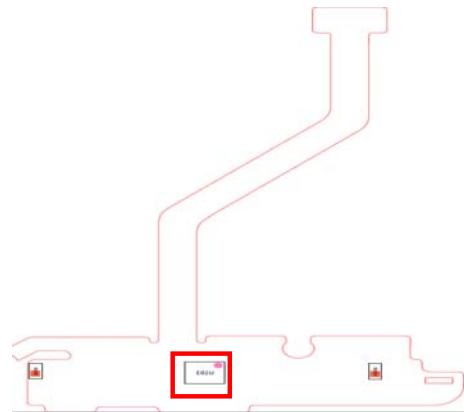
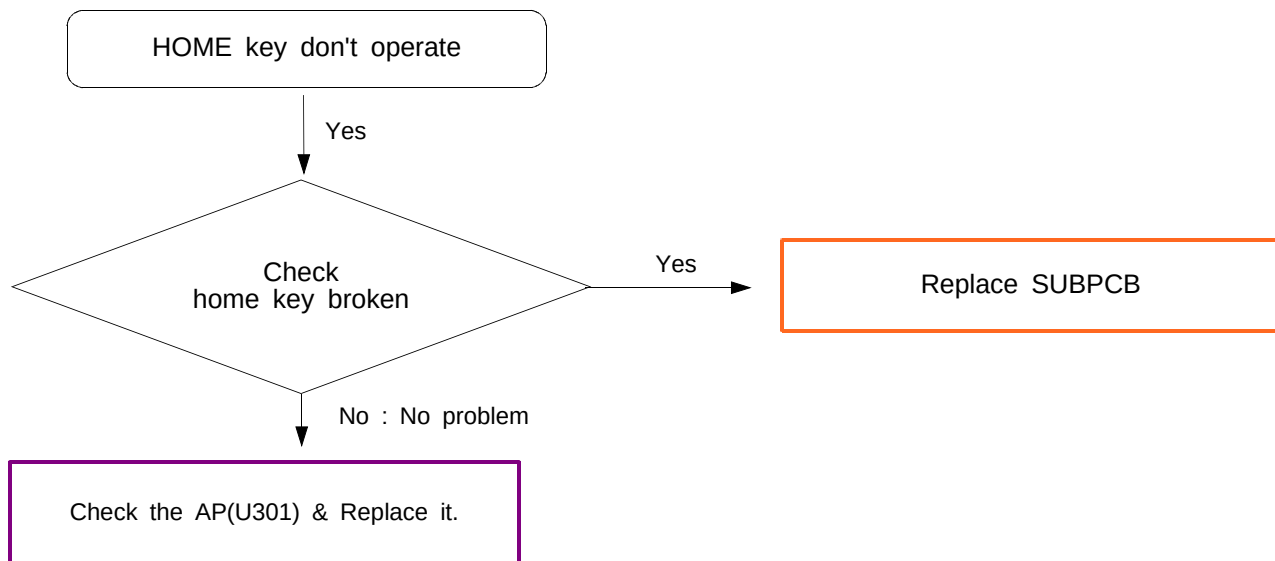


- power key GT-I8250

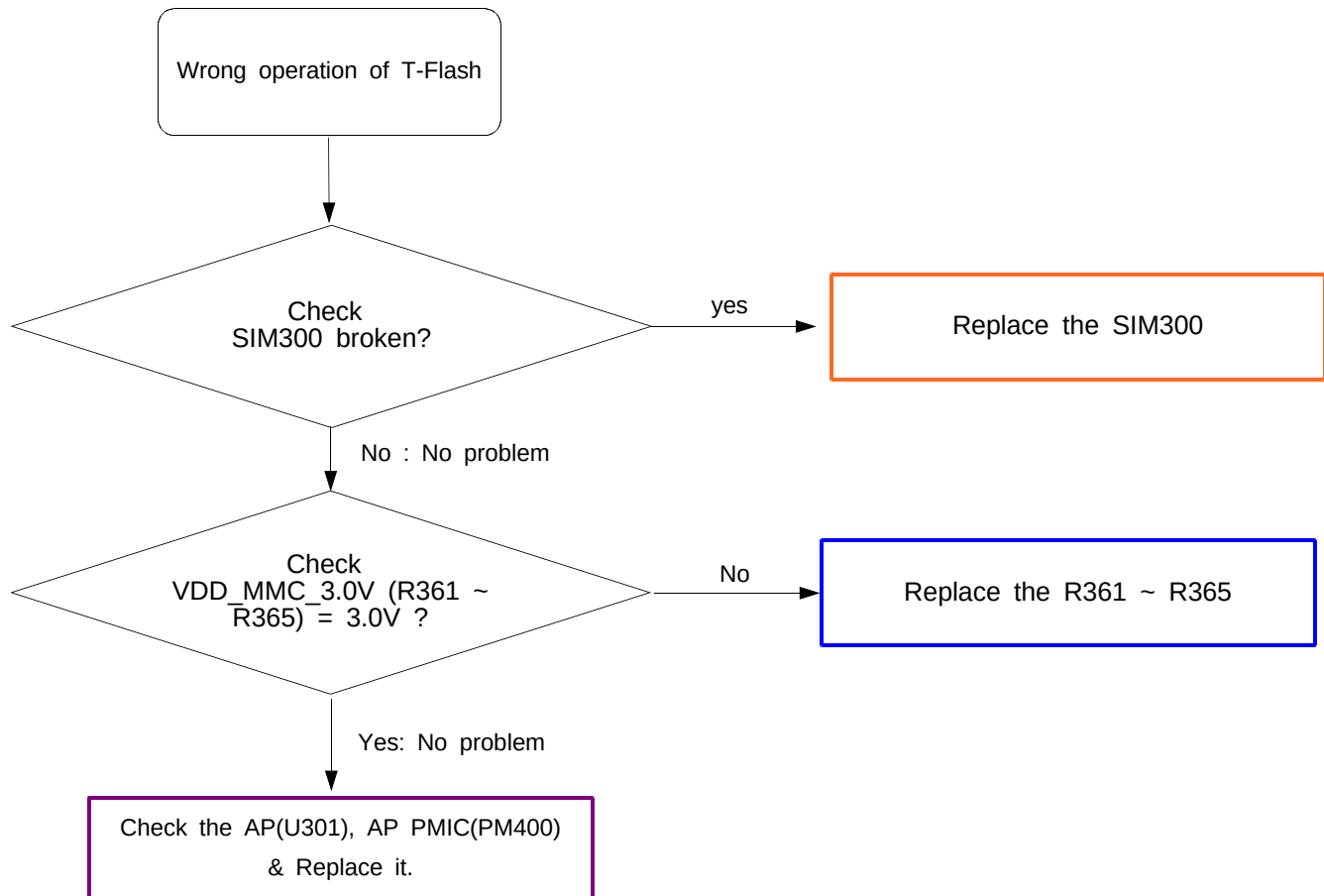


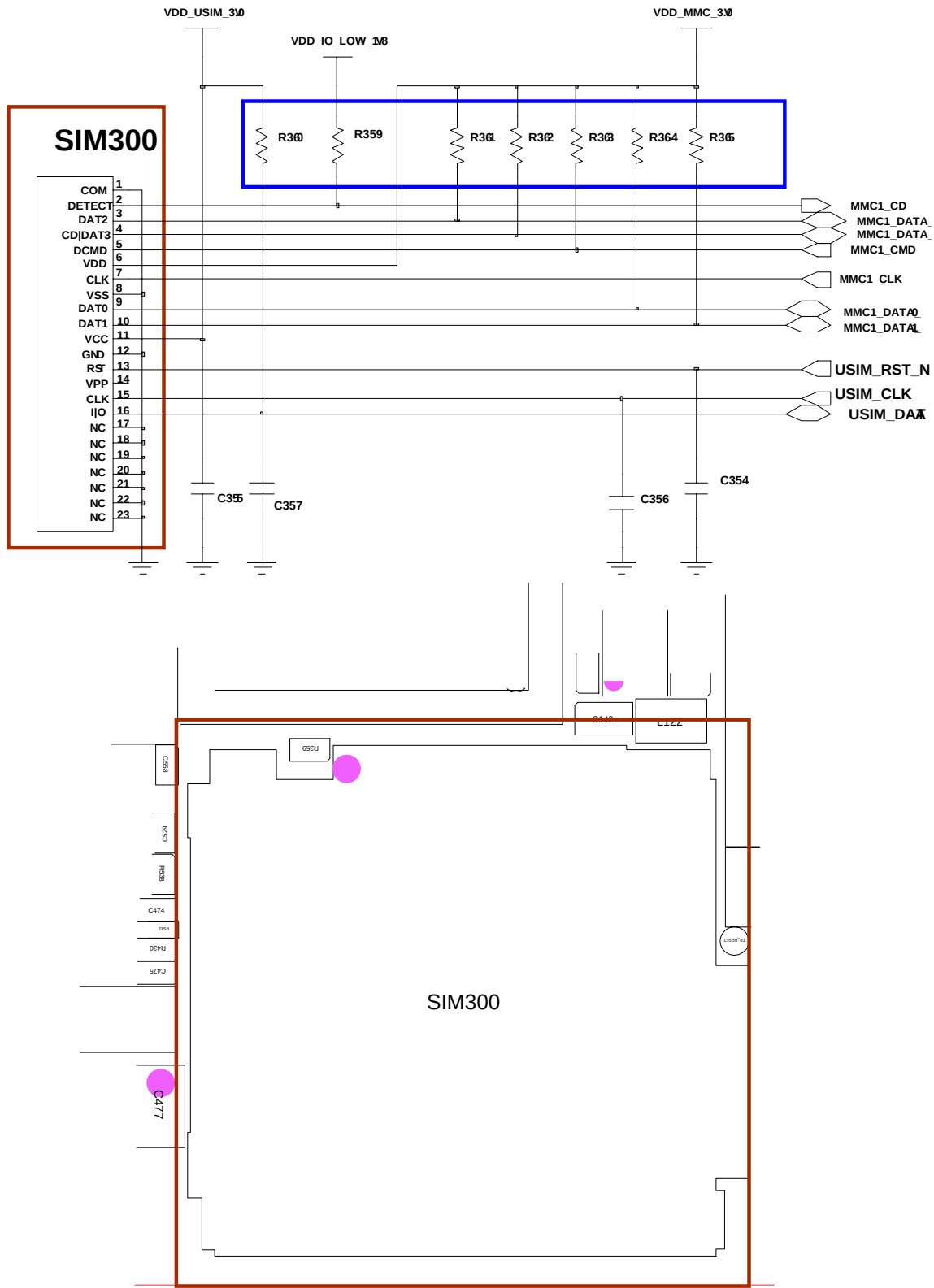


HOME key GT-i8250



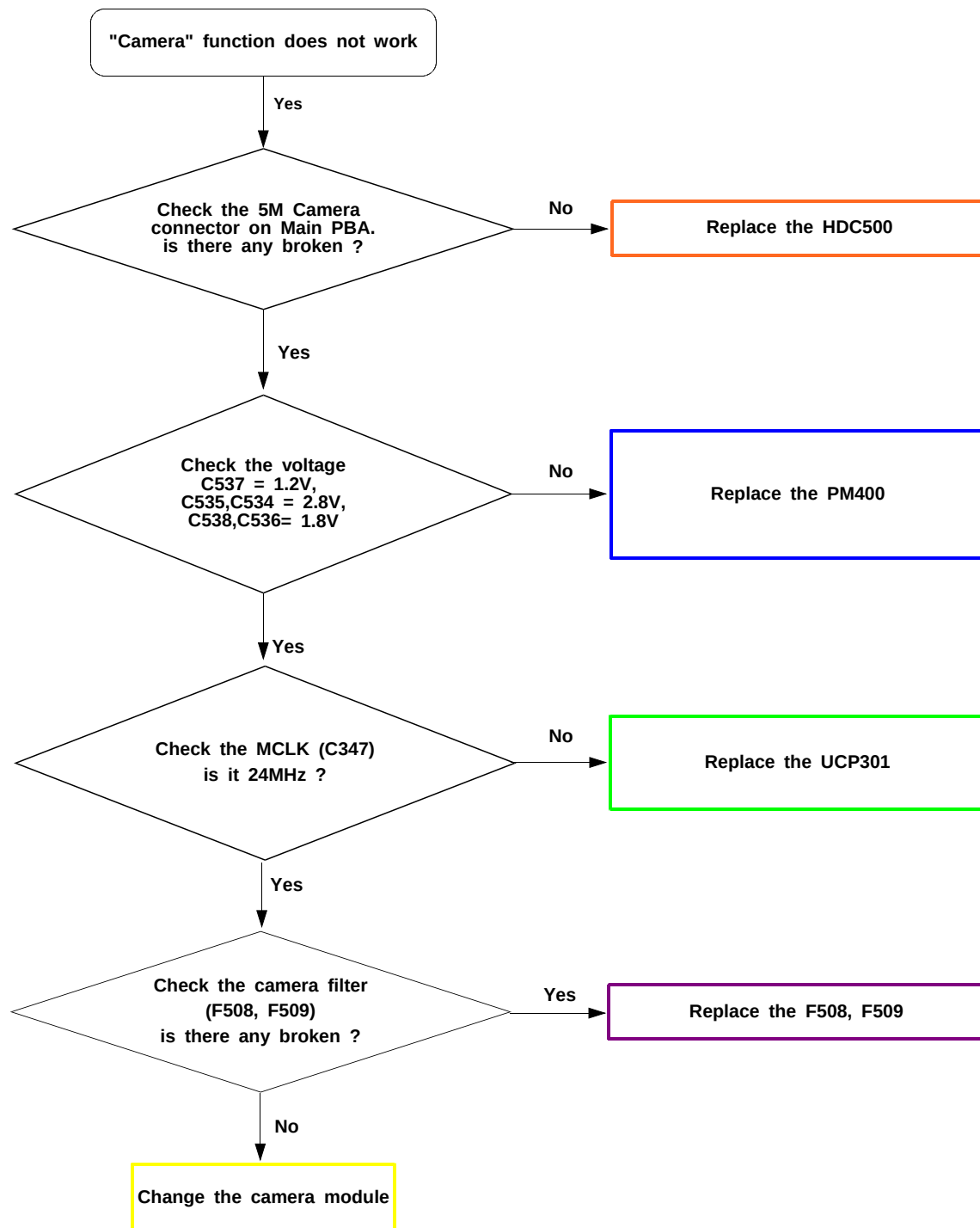
8-3-10. T-Flash GT-I8250



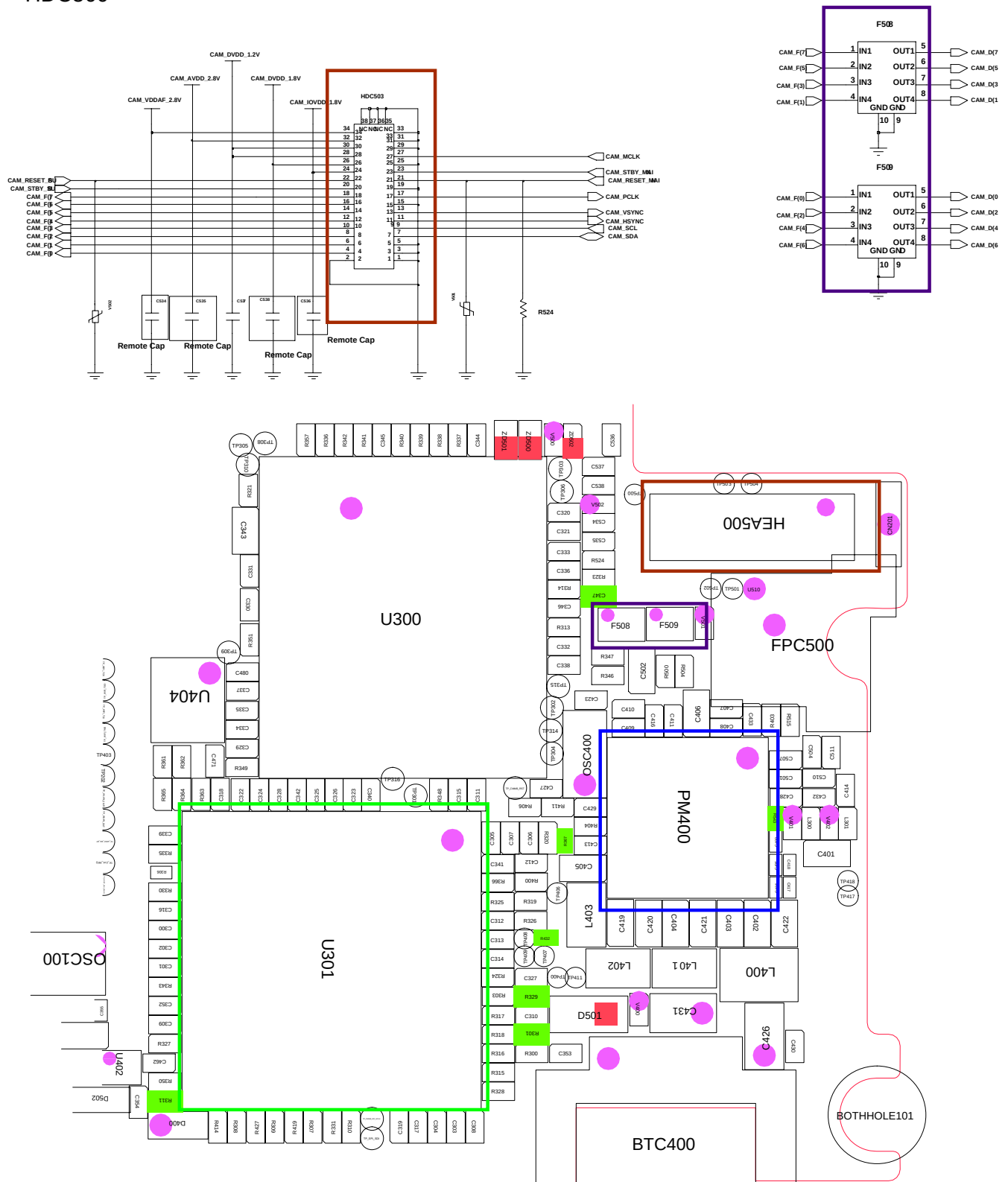


8-3-11. Camera I8250

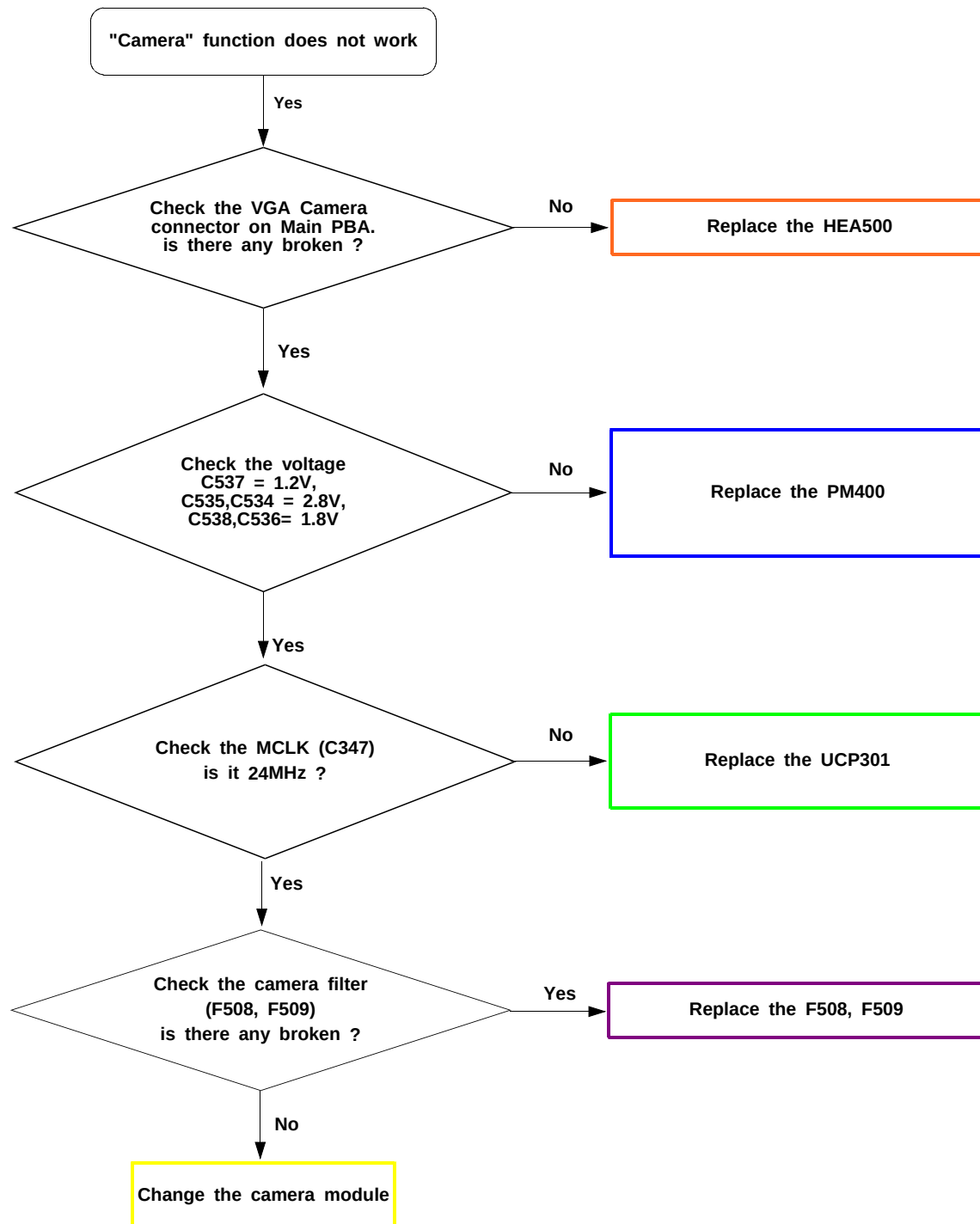
■ 5M Camera



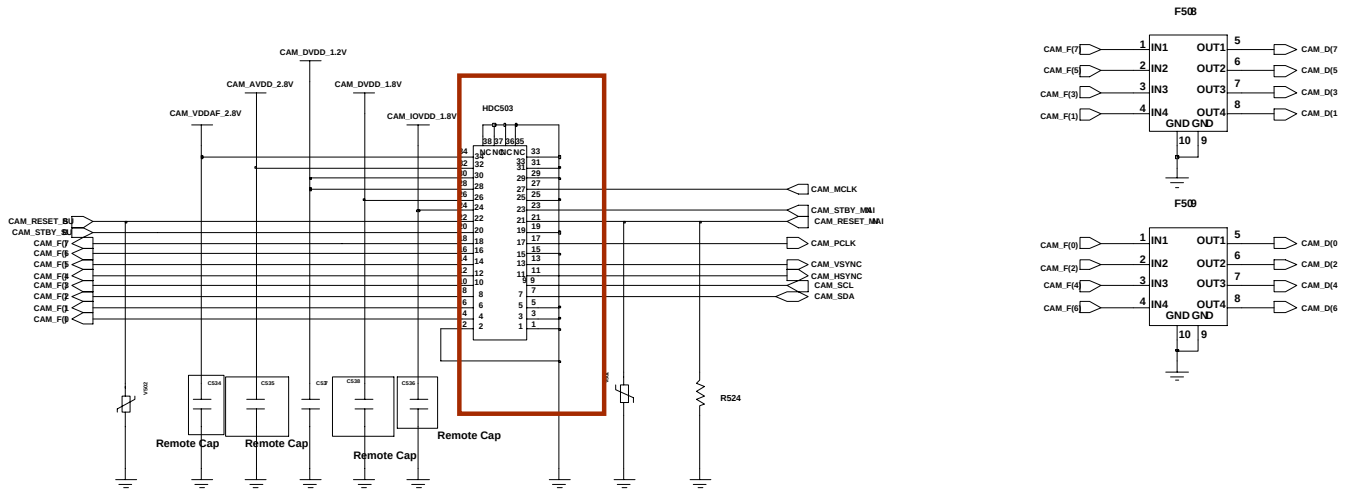
- HDC500



■ VGA Camera

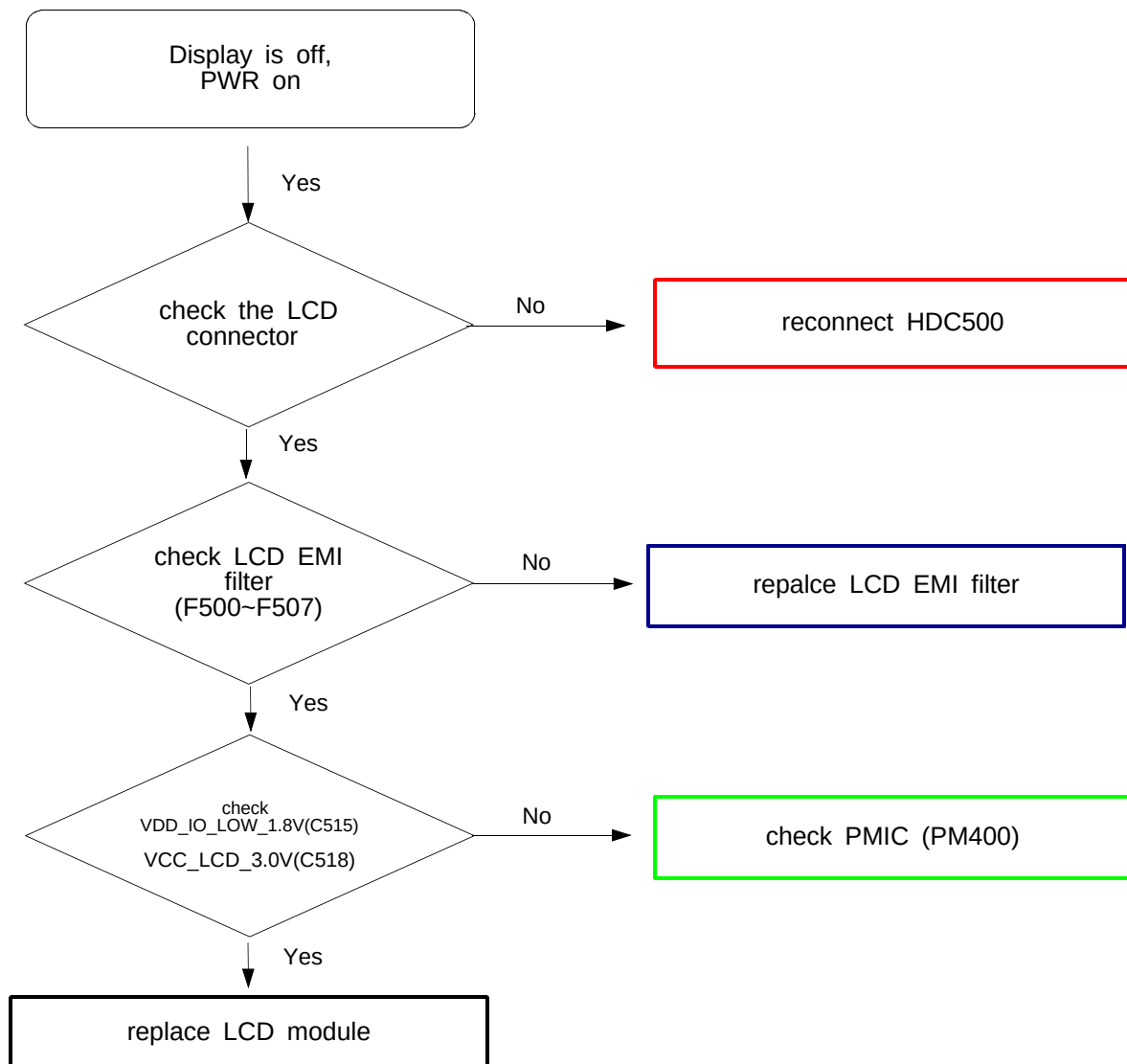


HDC500

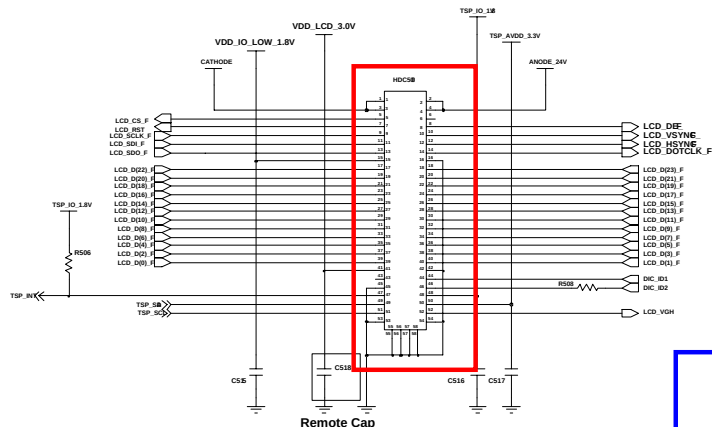


8-3-12. GT-I8250 LCD Working (TFT LCD,TSP)

- LCD

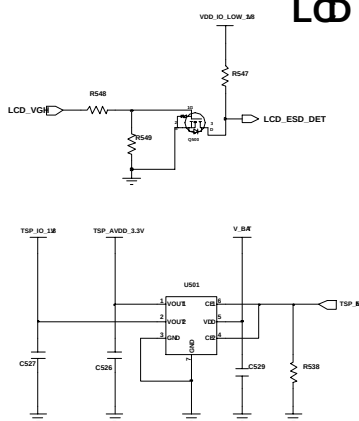


LCD connector, LCD EMI FILTER, LCD POWER

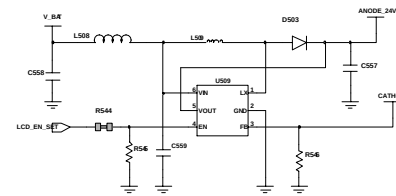


Remote Cap

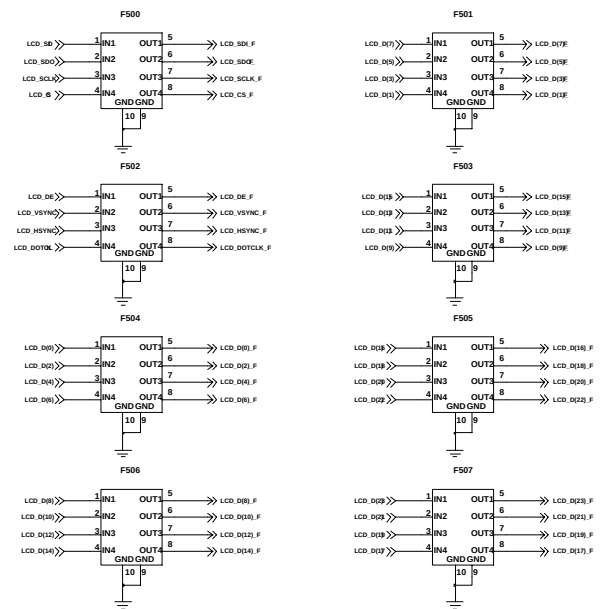
LO

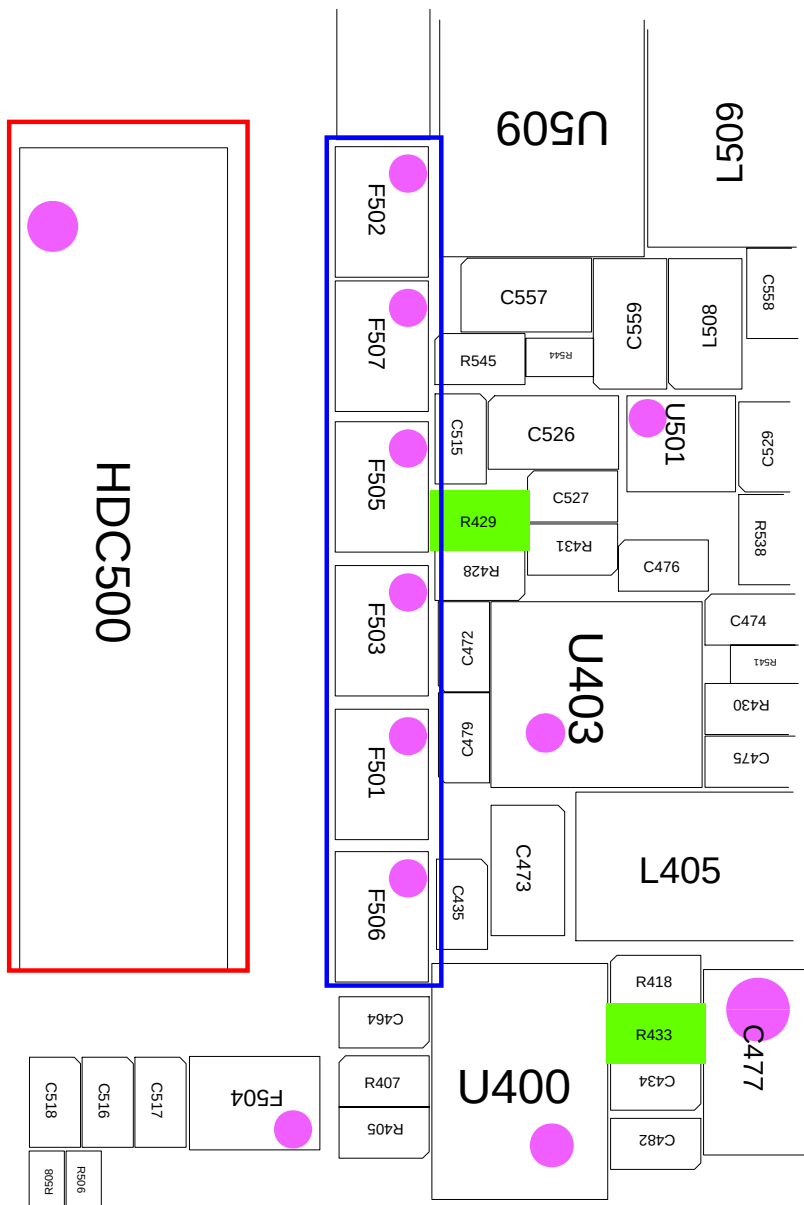


TSP PW

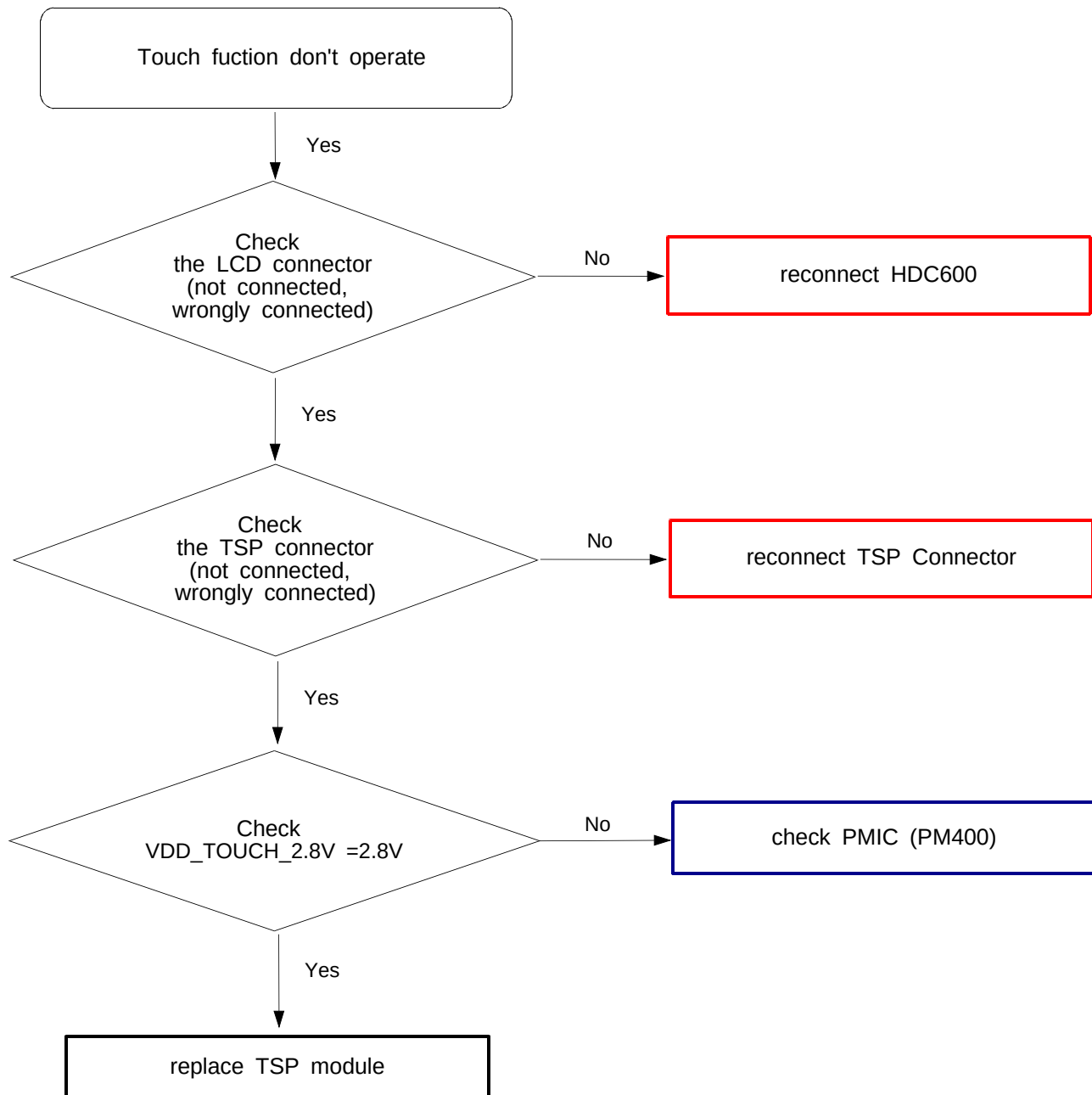


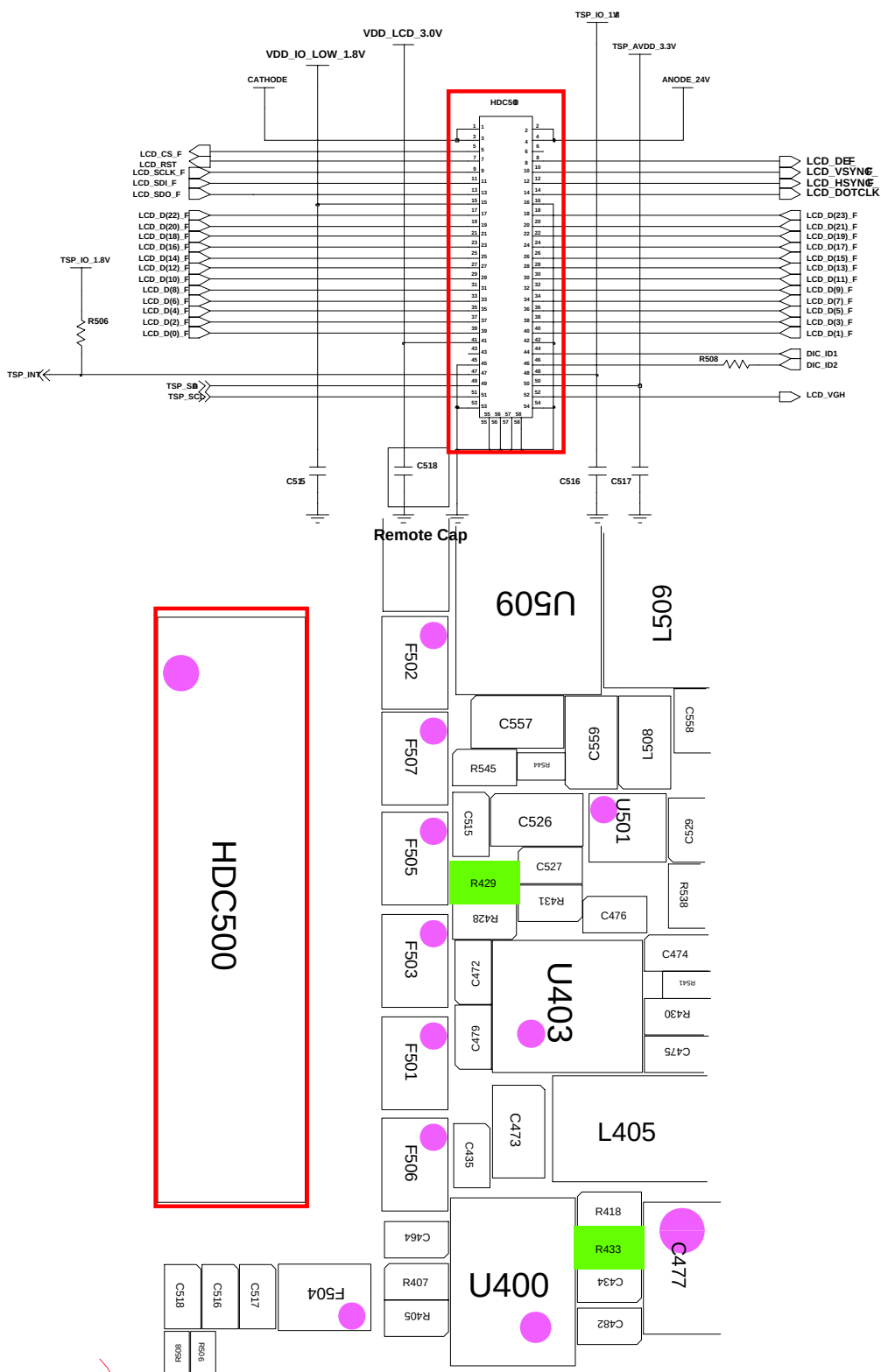
LCD Backlight Drive I





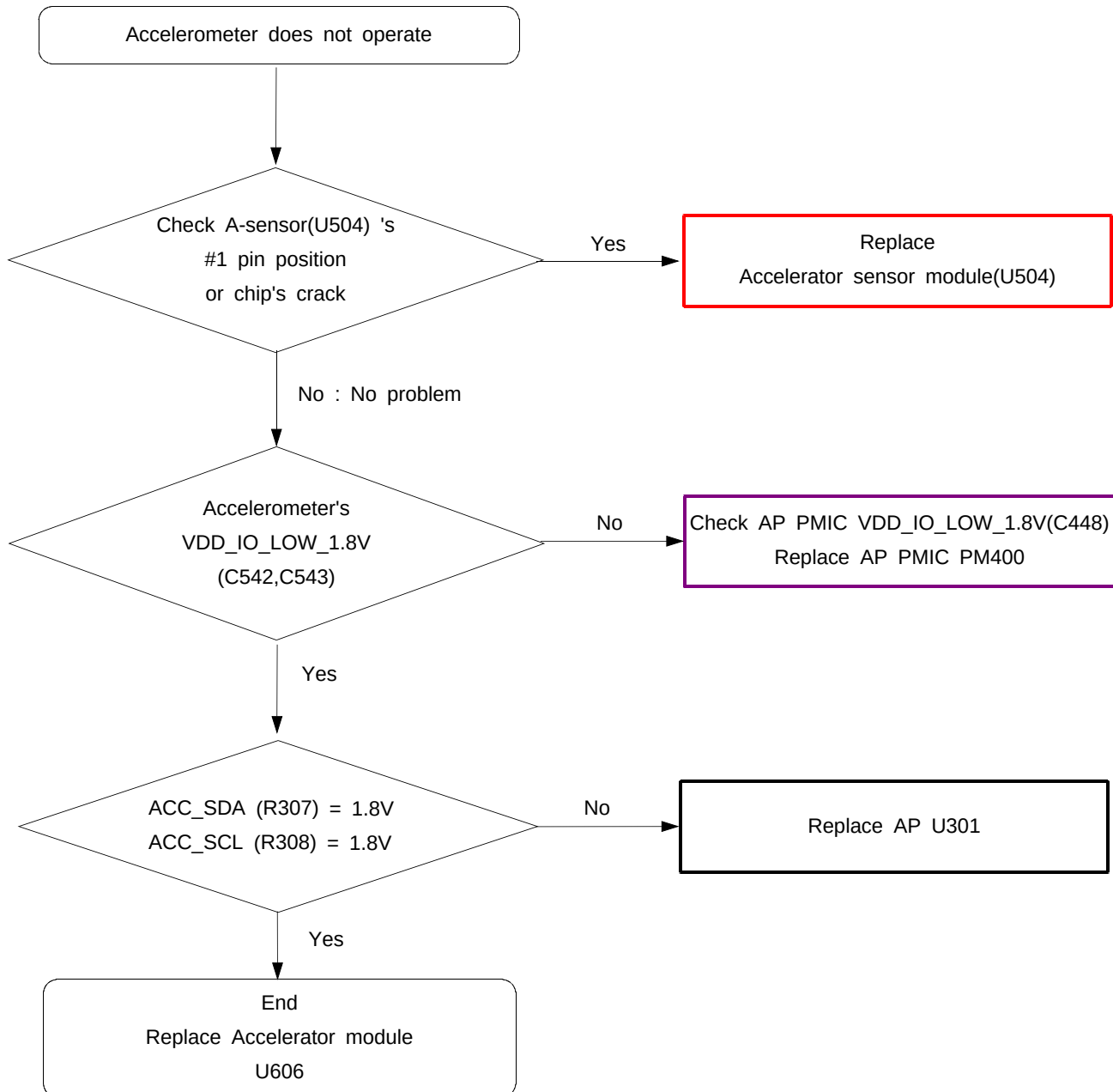
-TSP



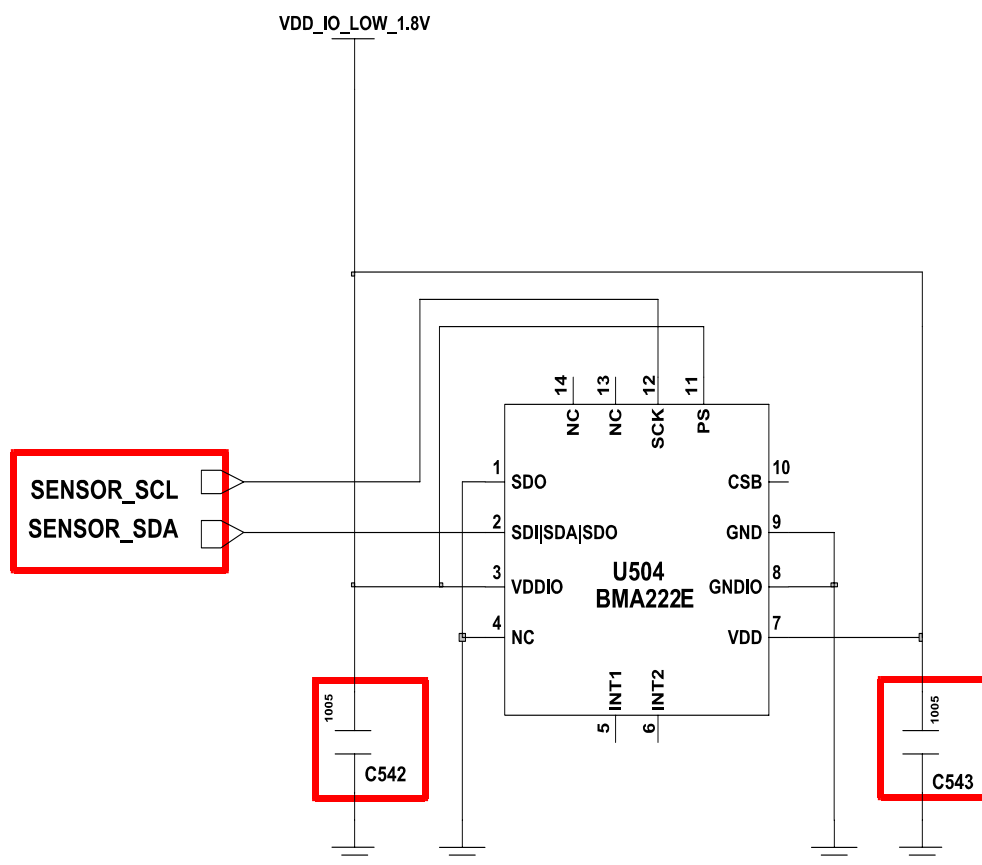
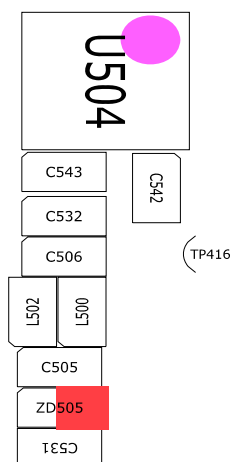


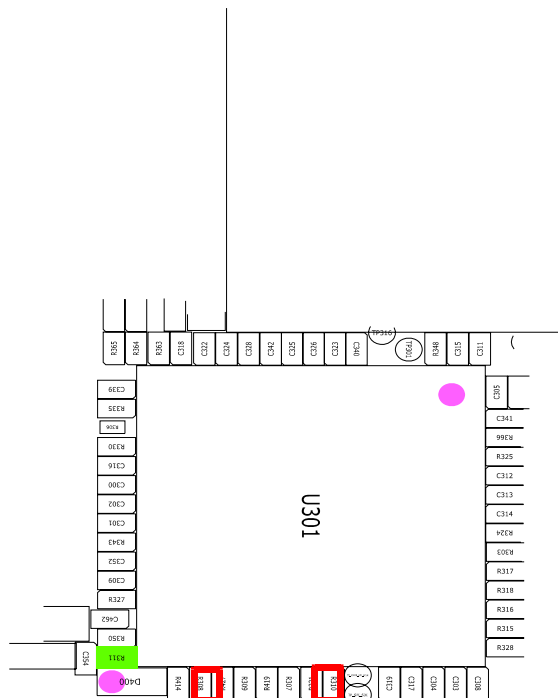
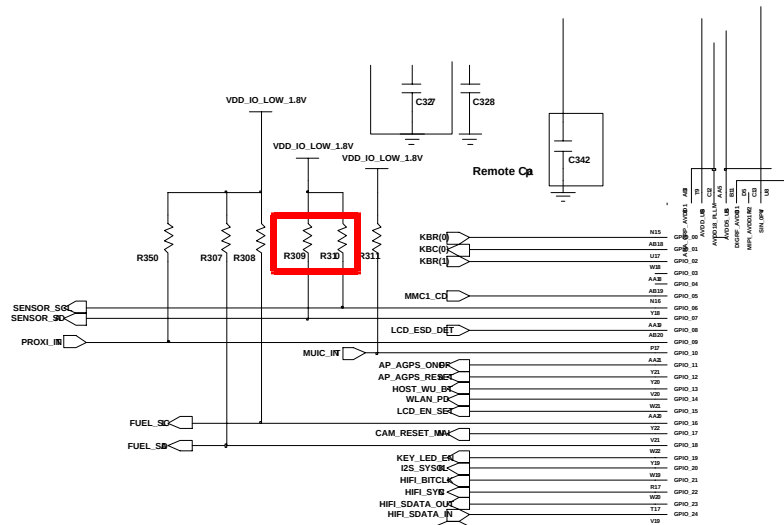
8-3-13. sensor function

- Accelerometer(3-Axis) GT-I8250

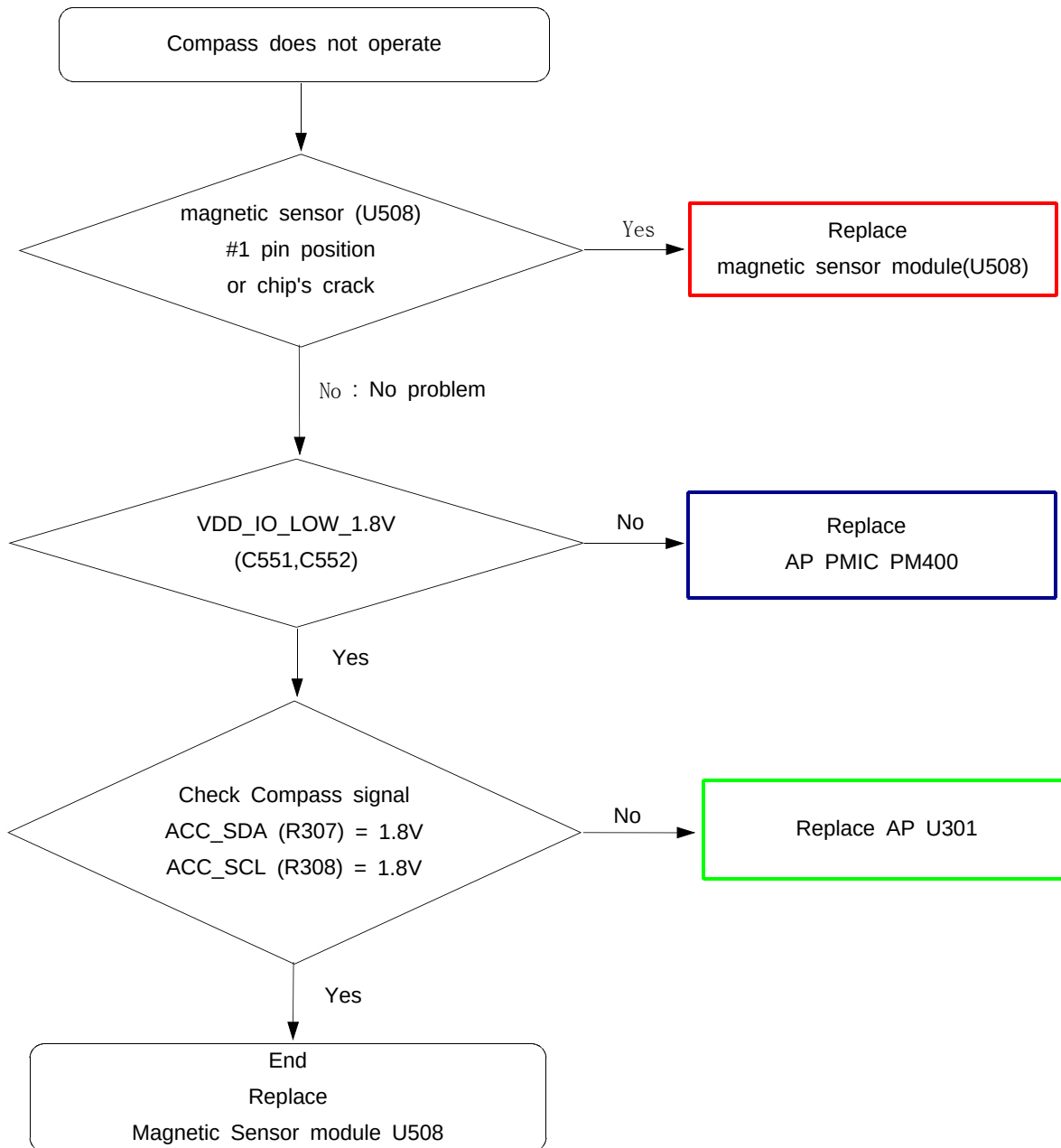


Accelerator sensor(U504) 's #1 pin position

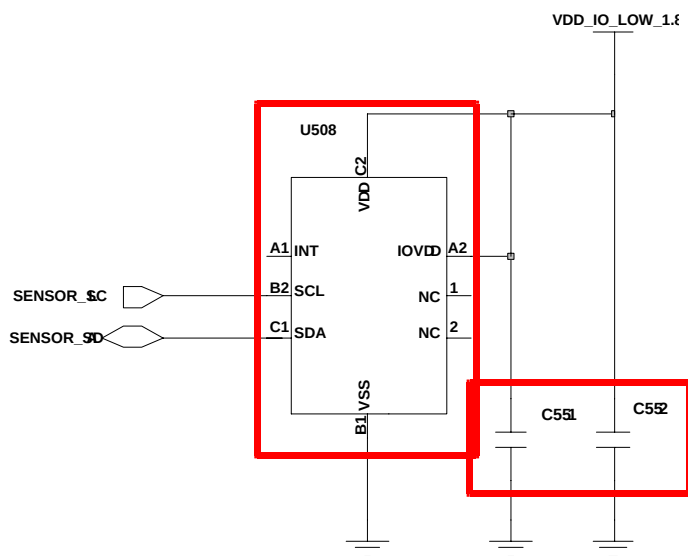
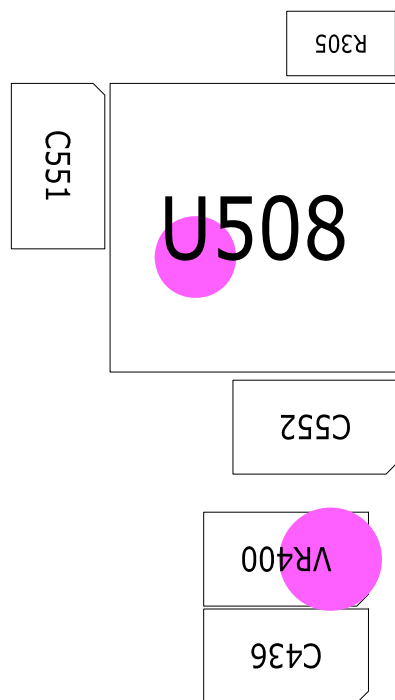


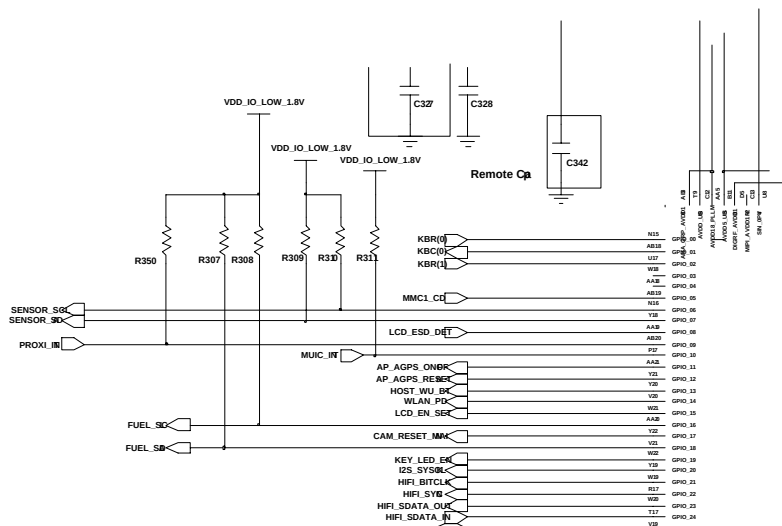
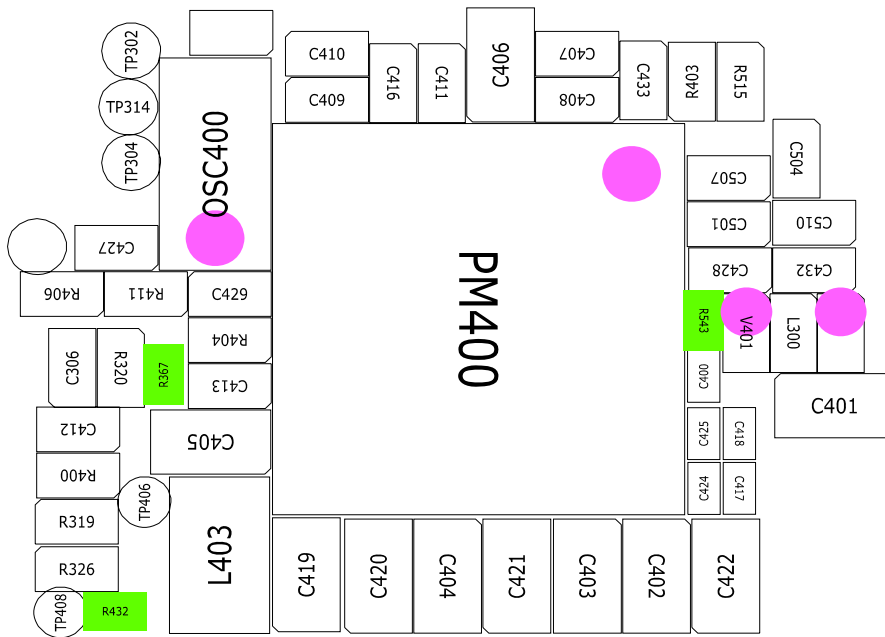


- Magnetic Sensor (U508) GT-I8250

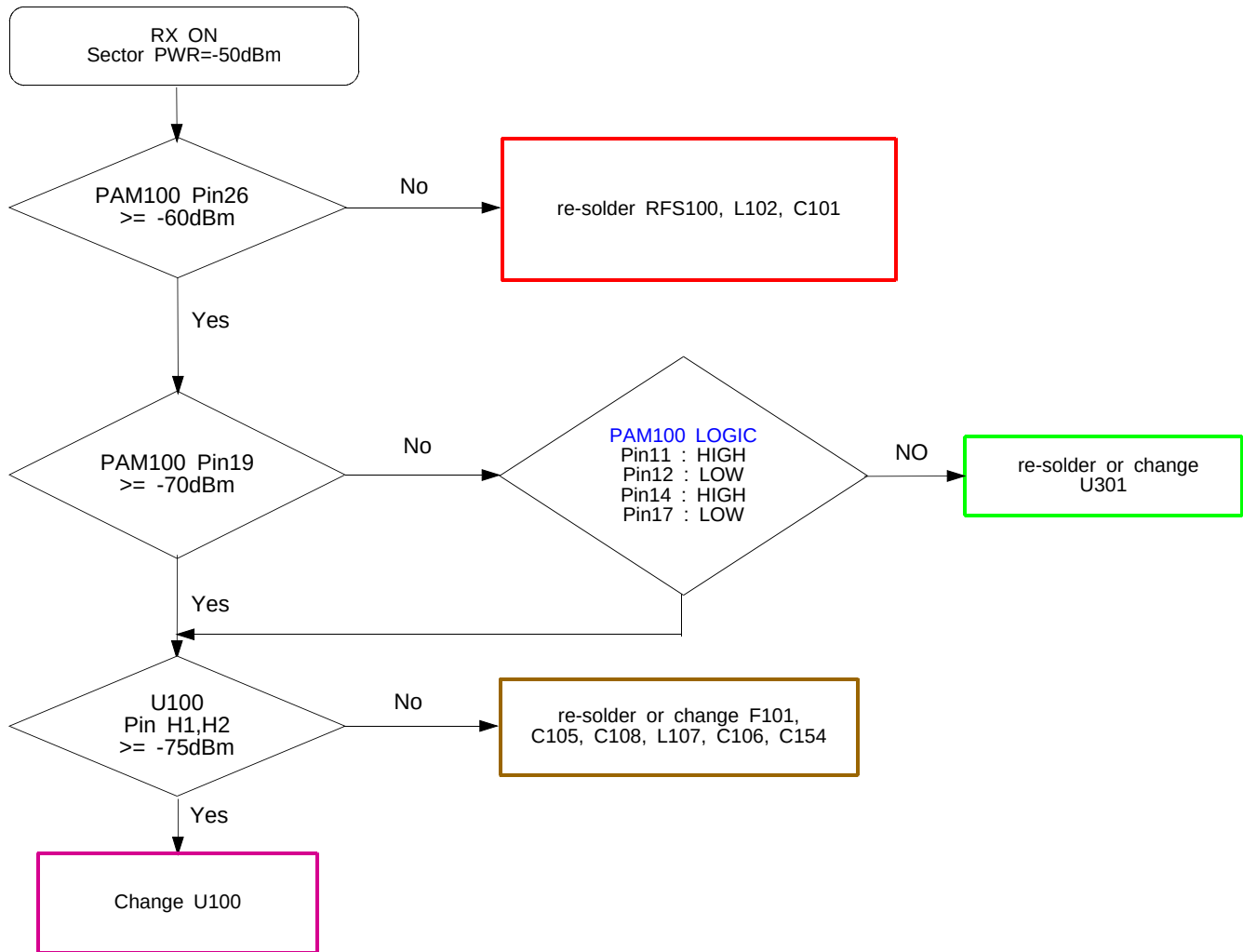


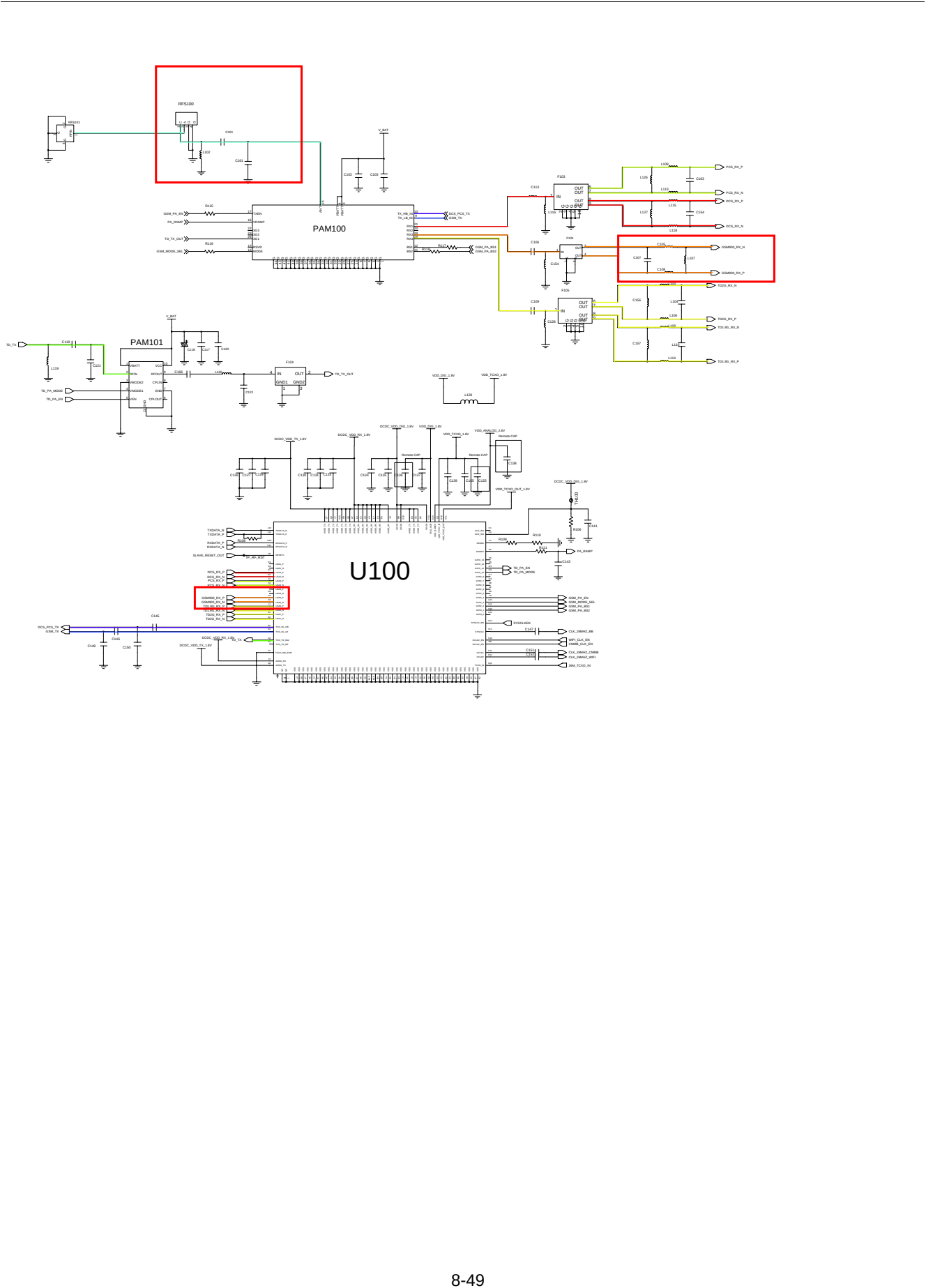
Magnetic sensor (U508)





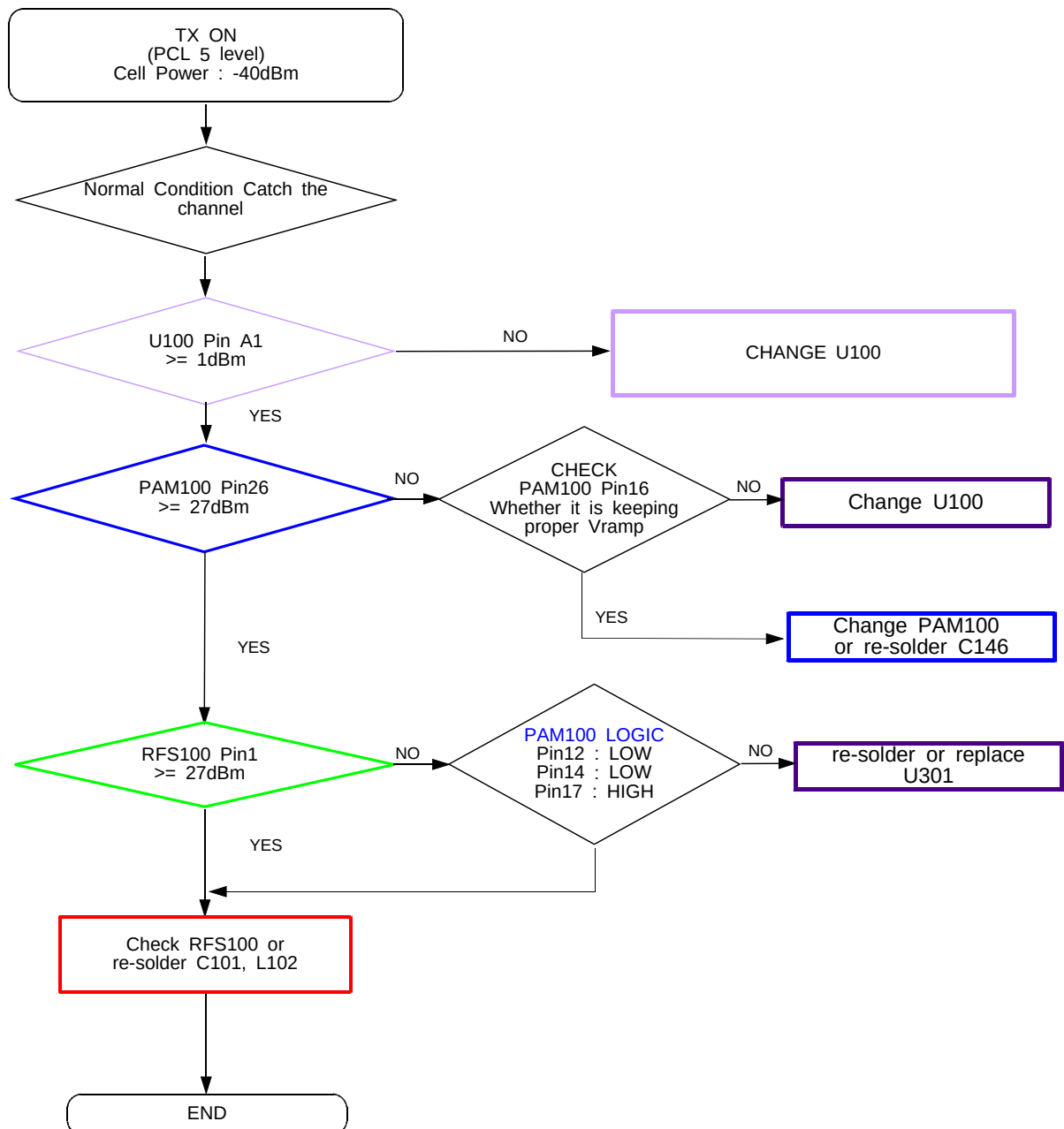
3-14. GSM900 Receiver I8250

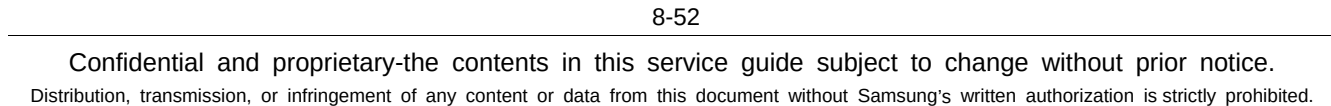


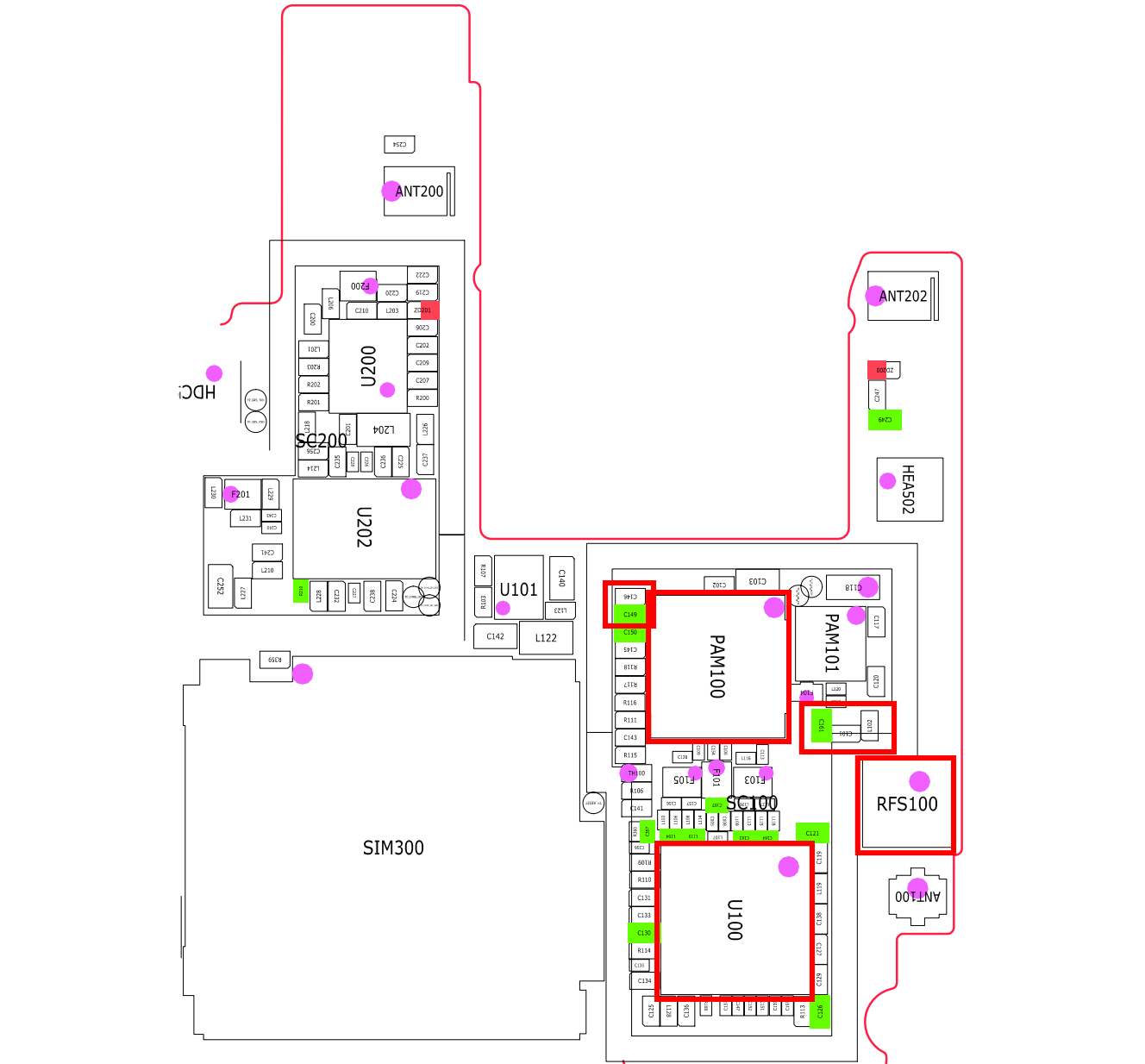




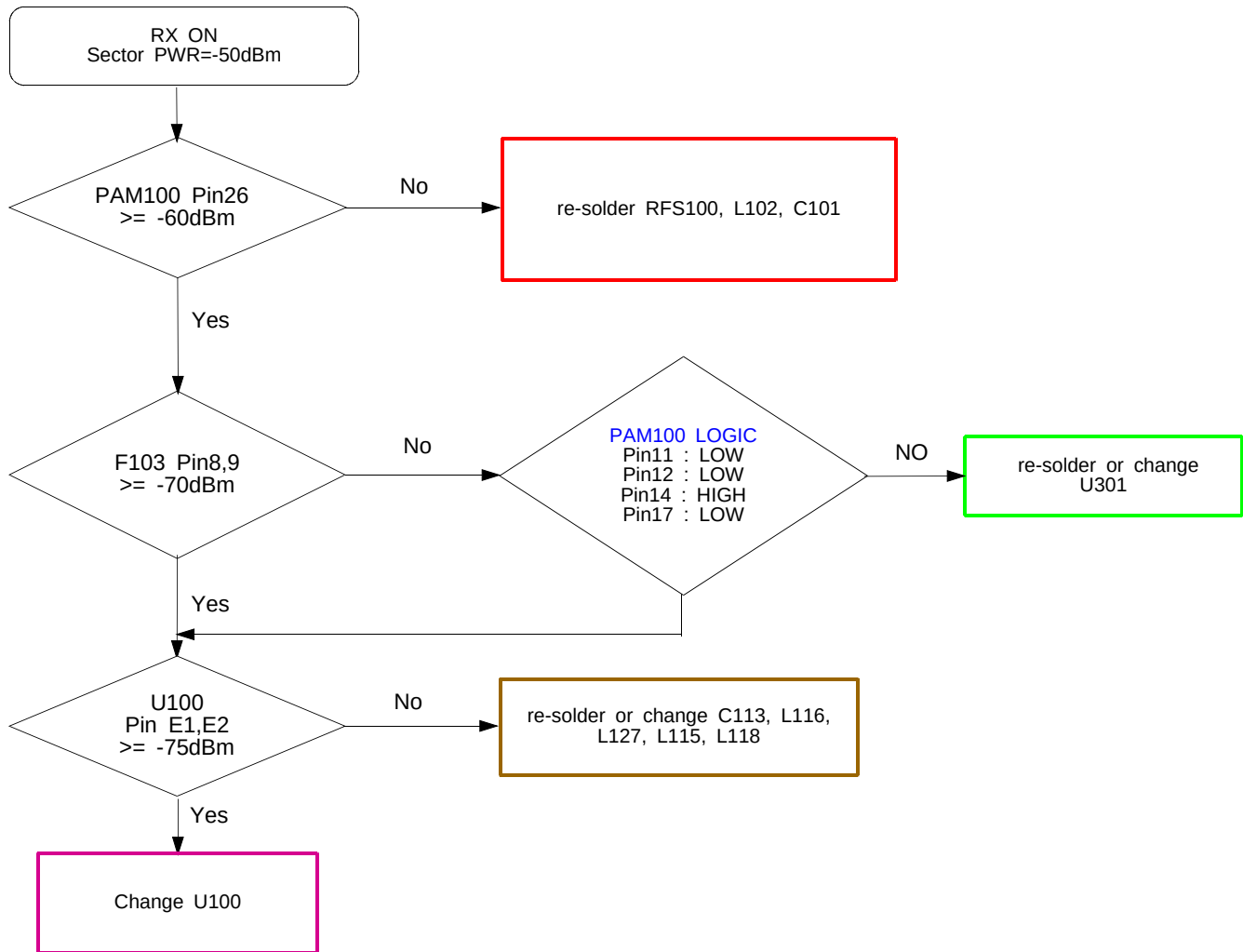
8-3-15. GSM900 Transmitter

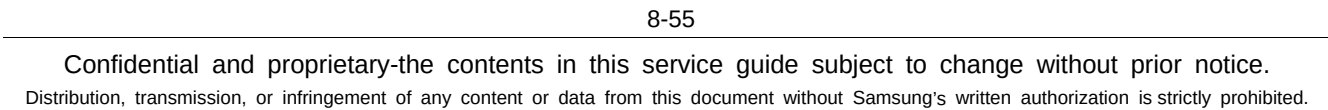


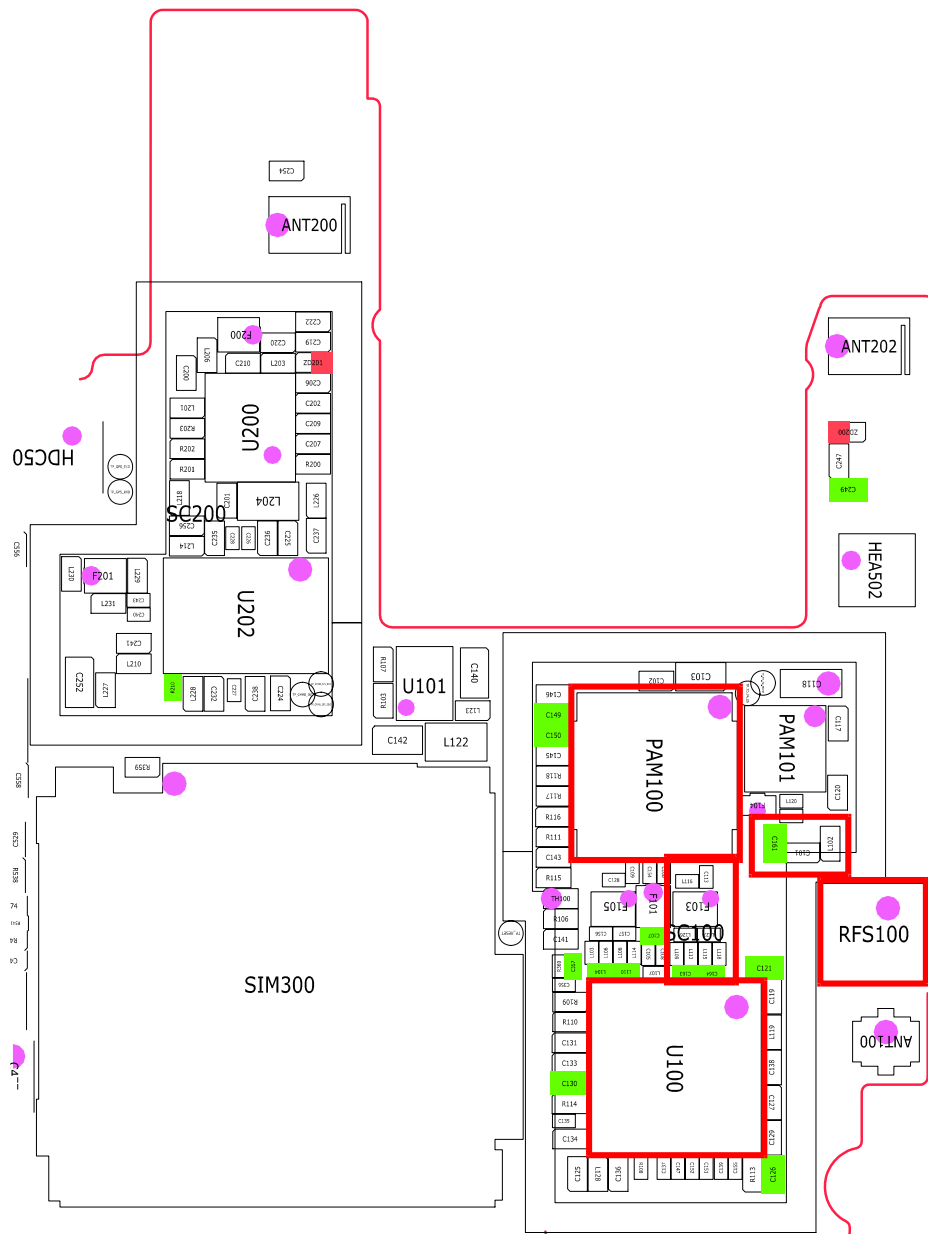




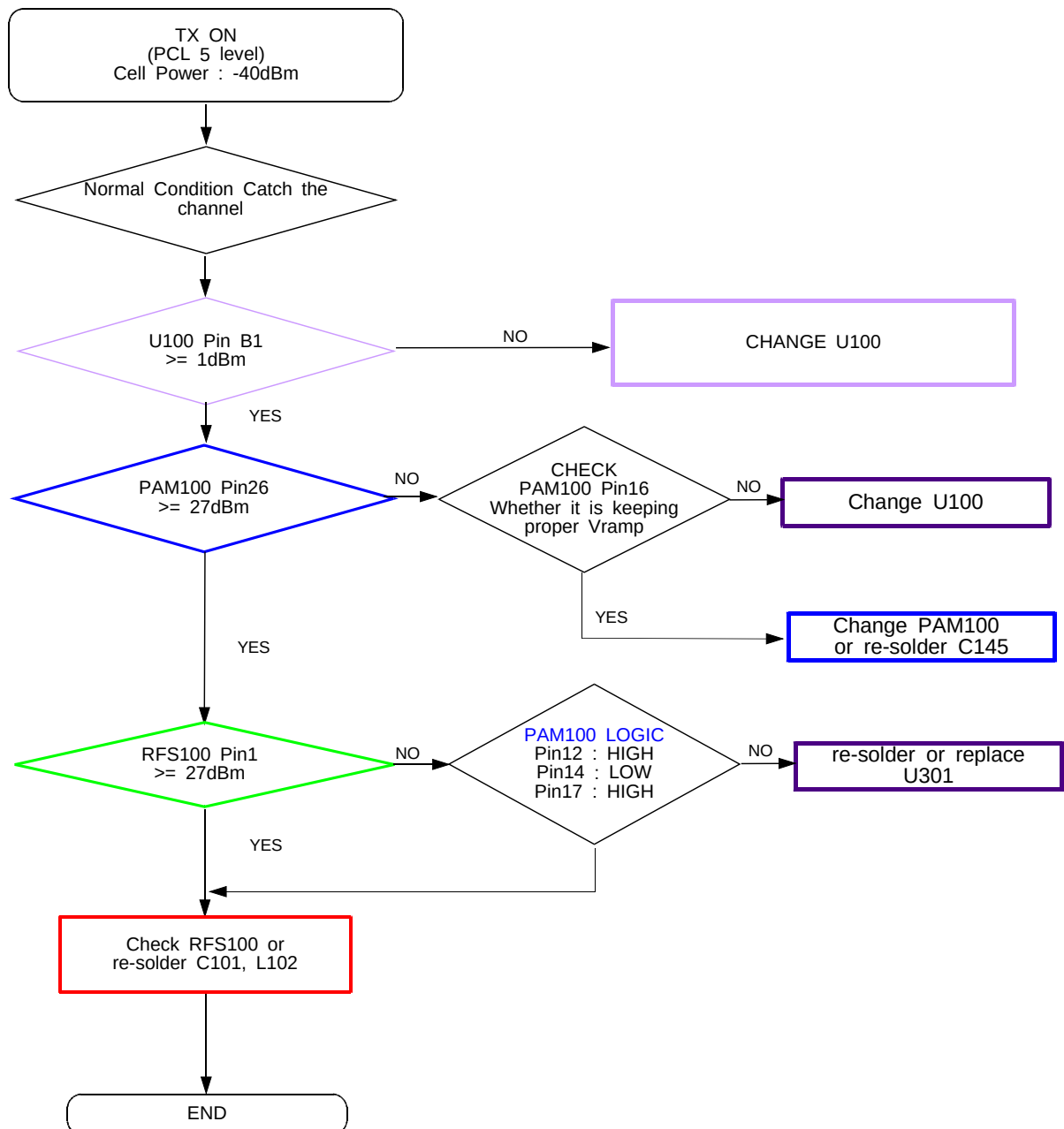
8-3-16. GSM1800 Receiver

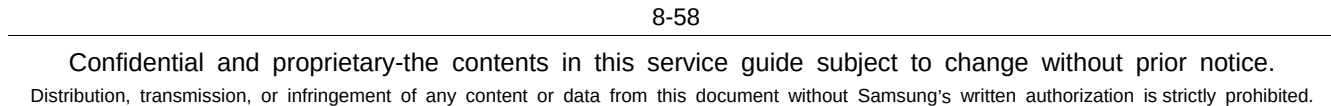






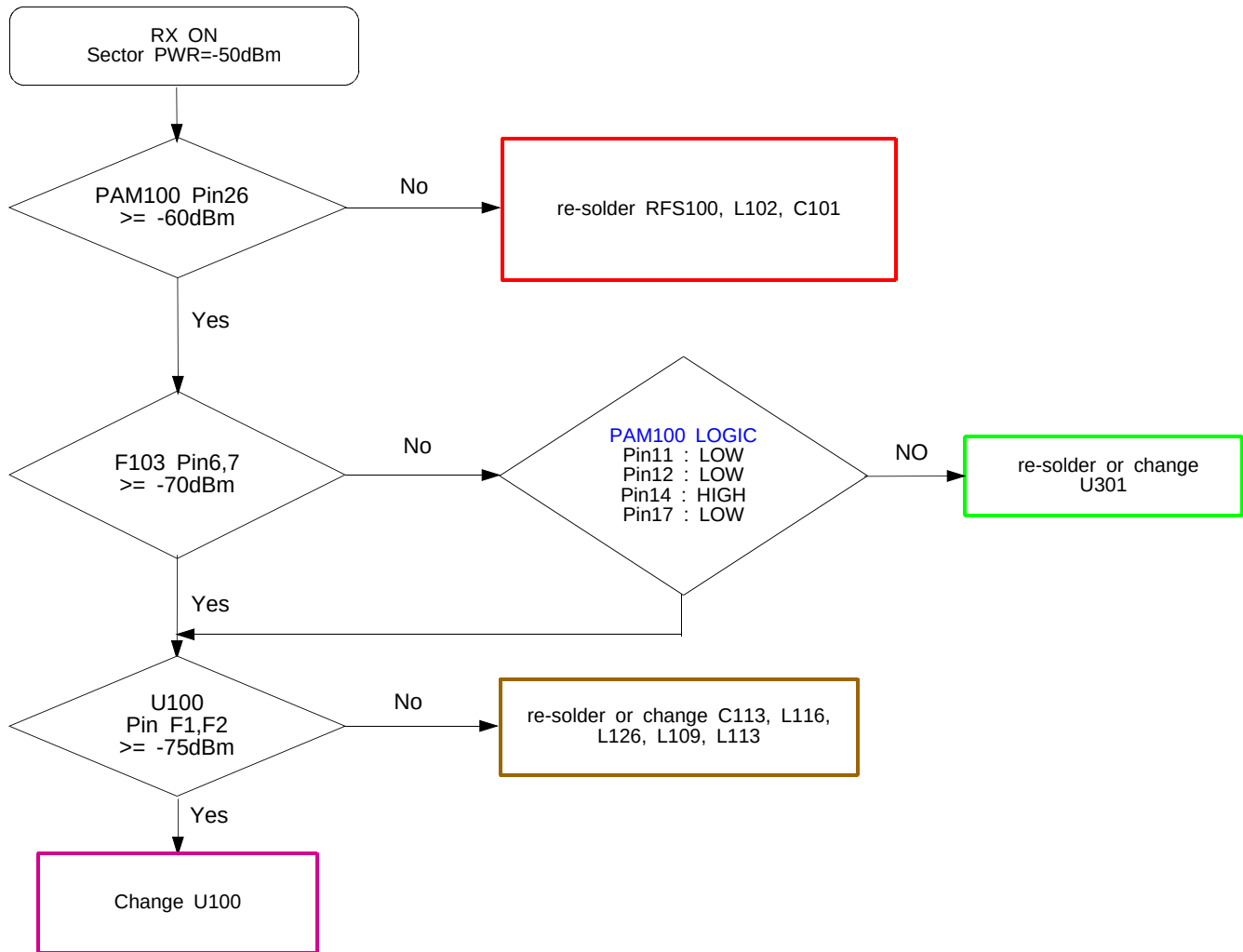
8-3-17. GSM1800 Transmitter

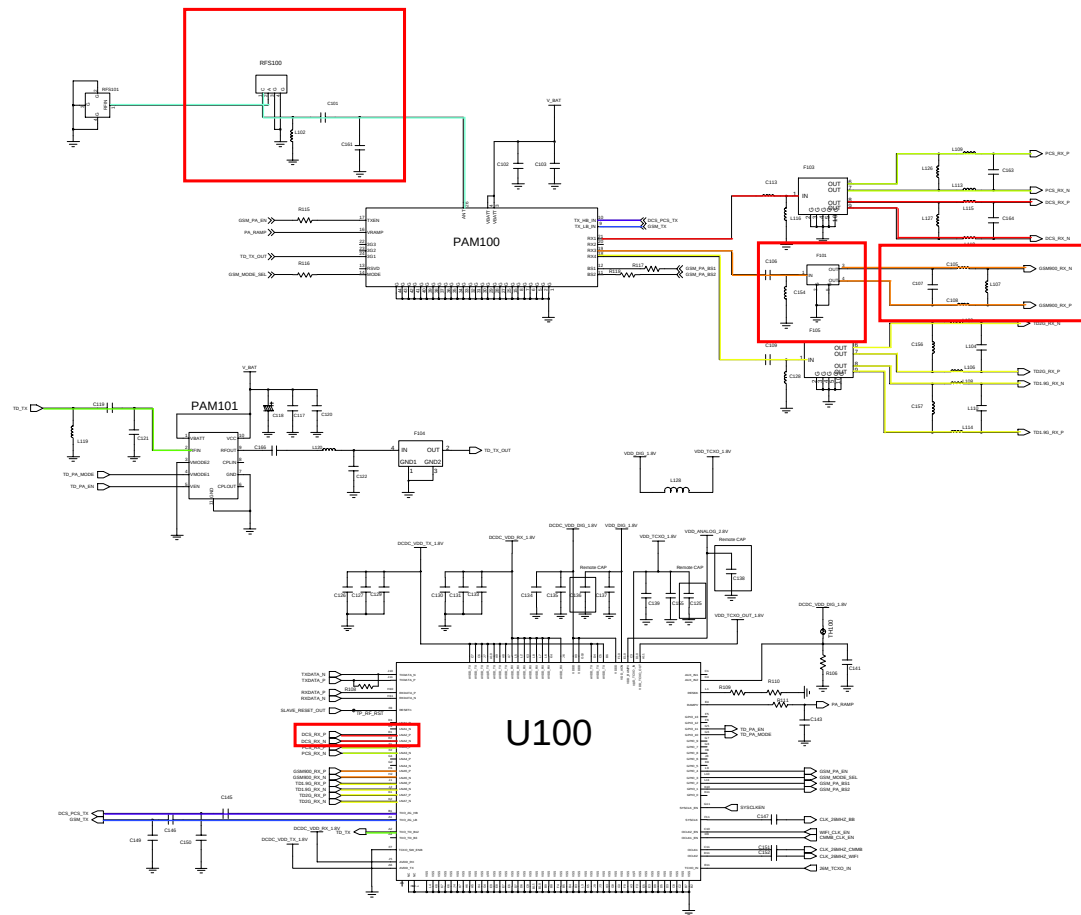


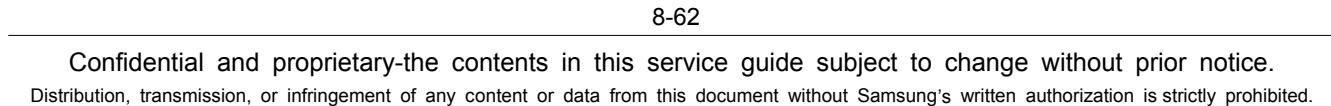




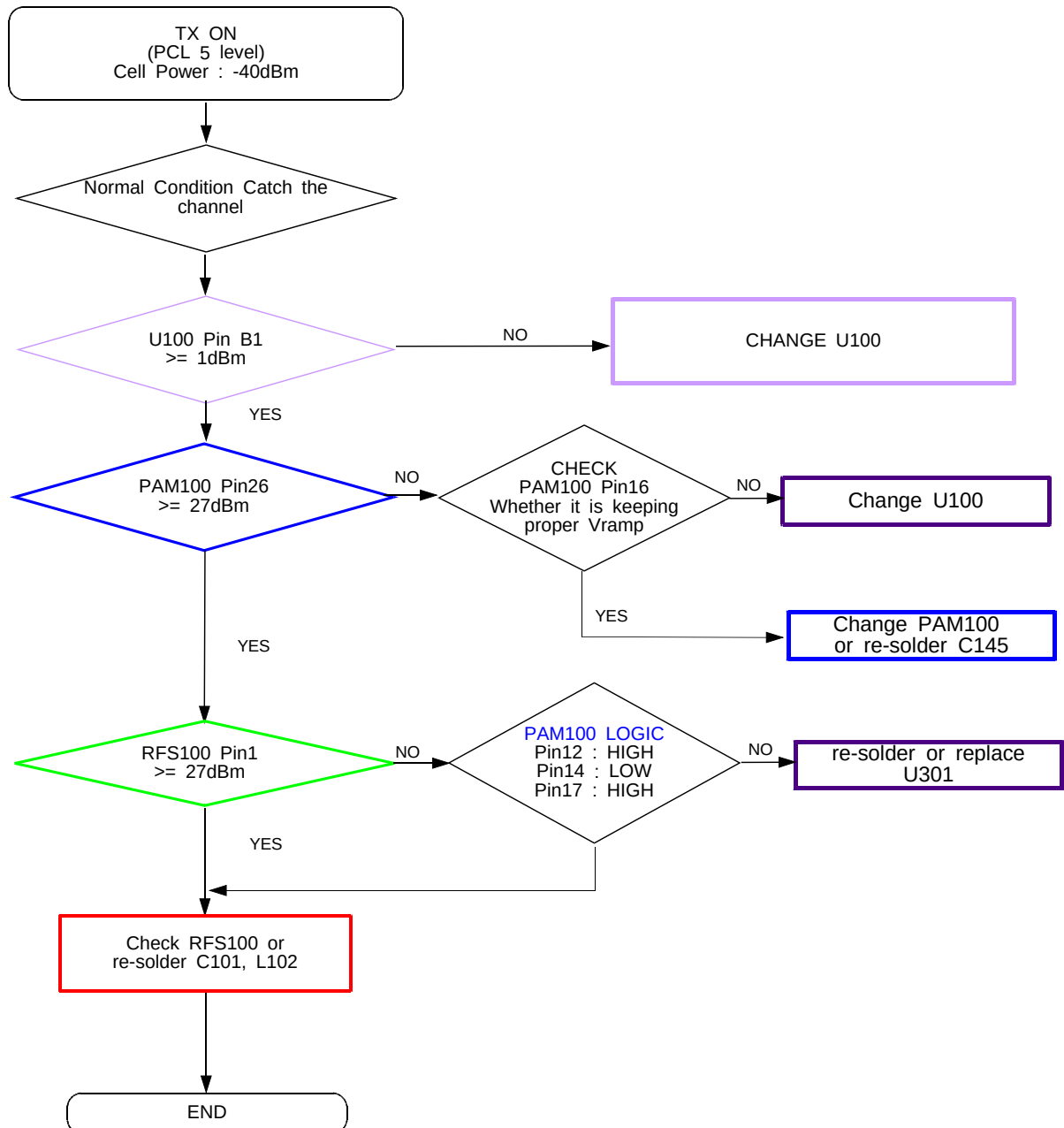
8-3-18. GSM1900 Receiver

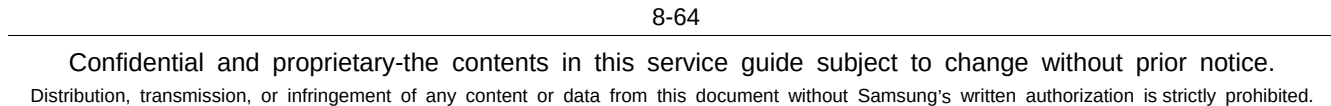






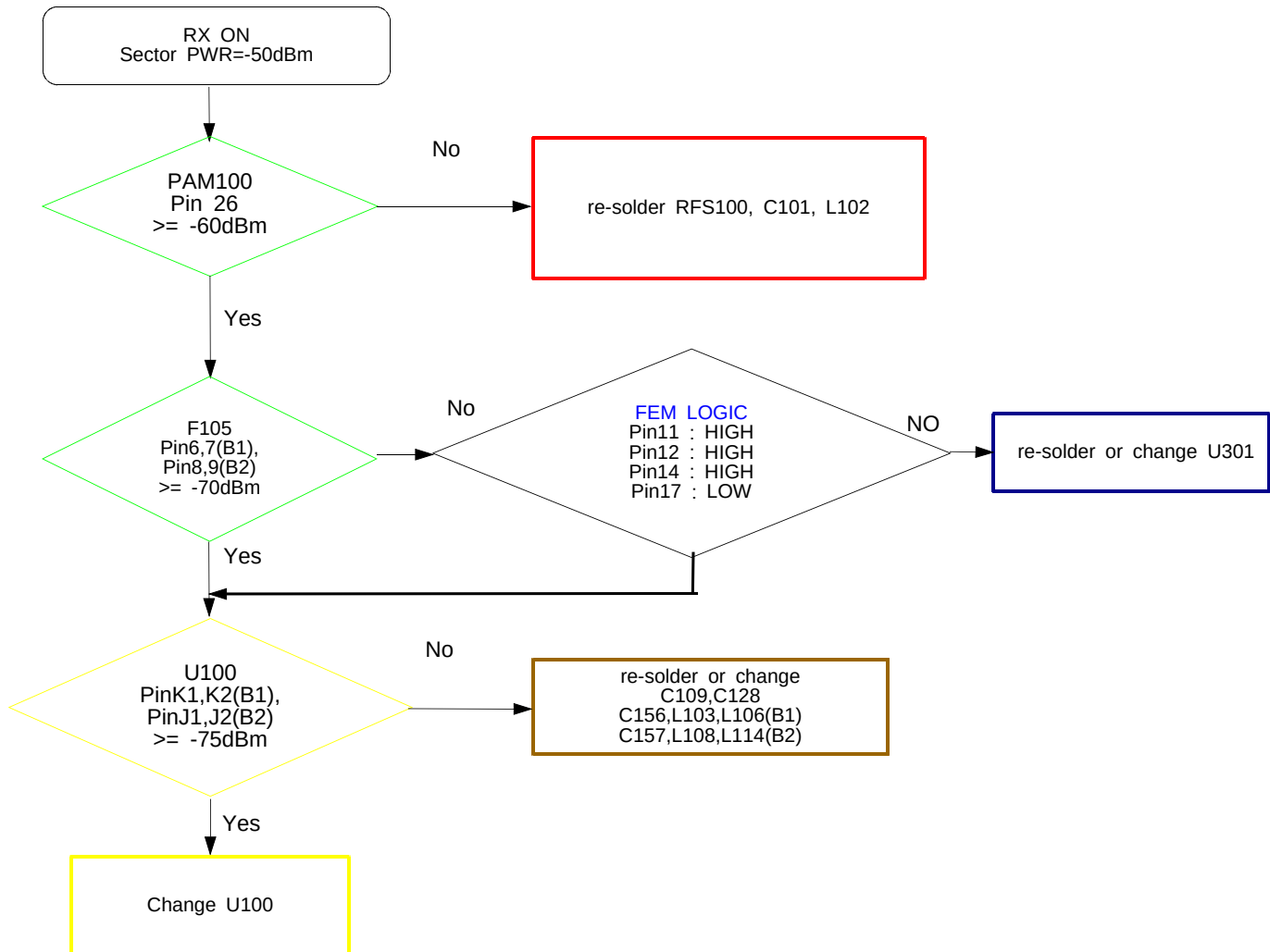
8-3-19. GSM1900 Transmitter

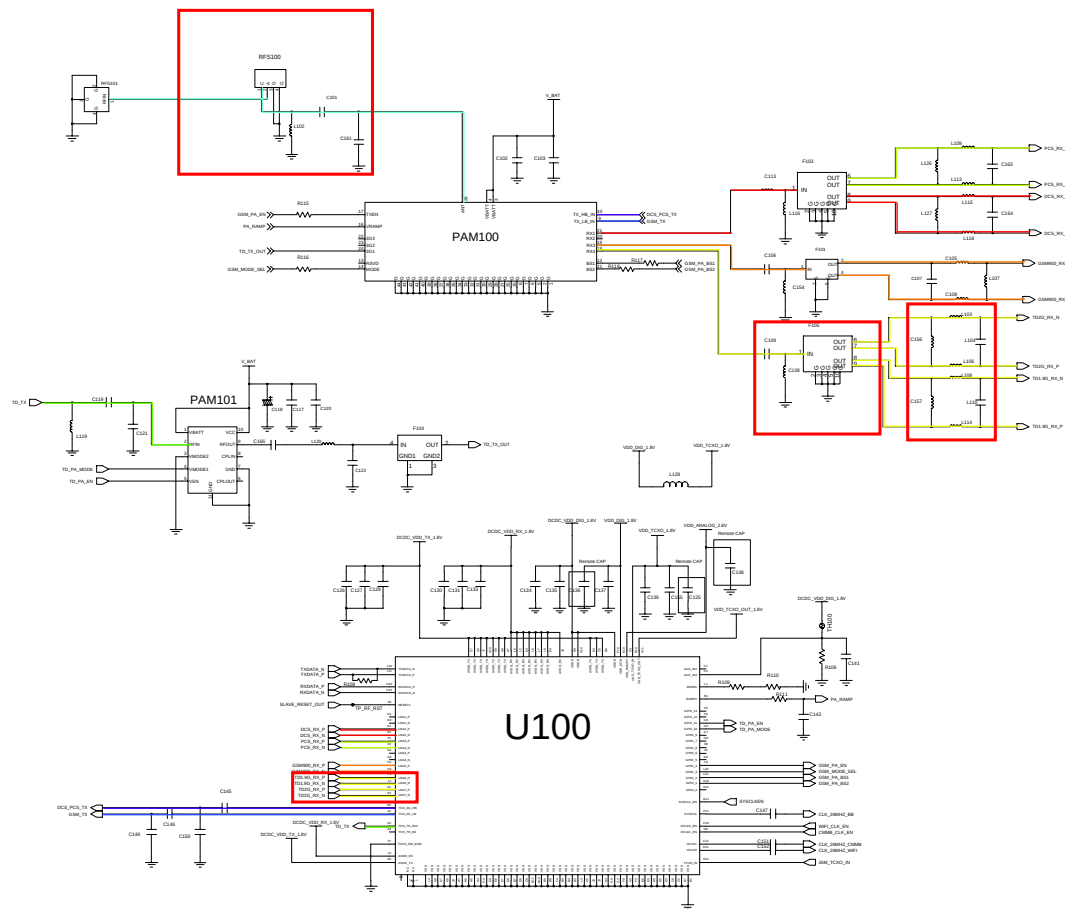






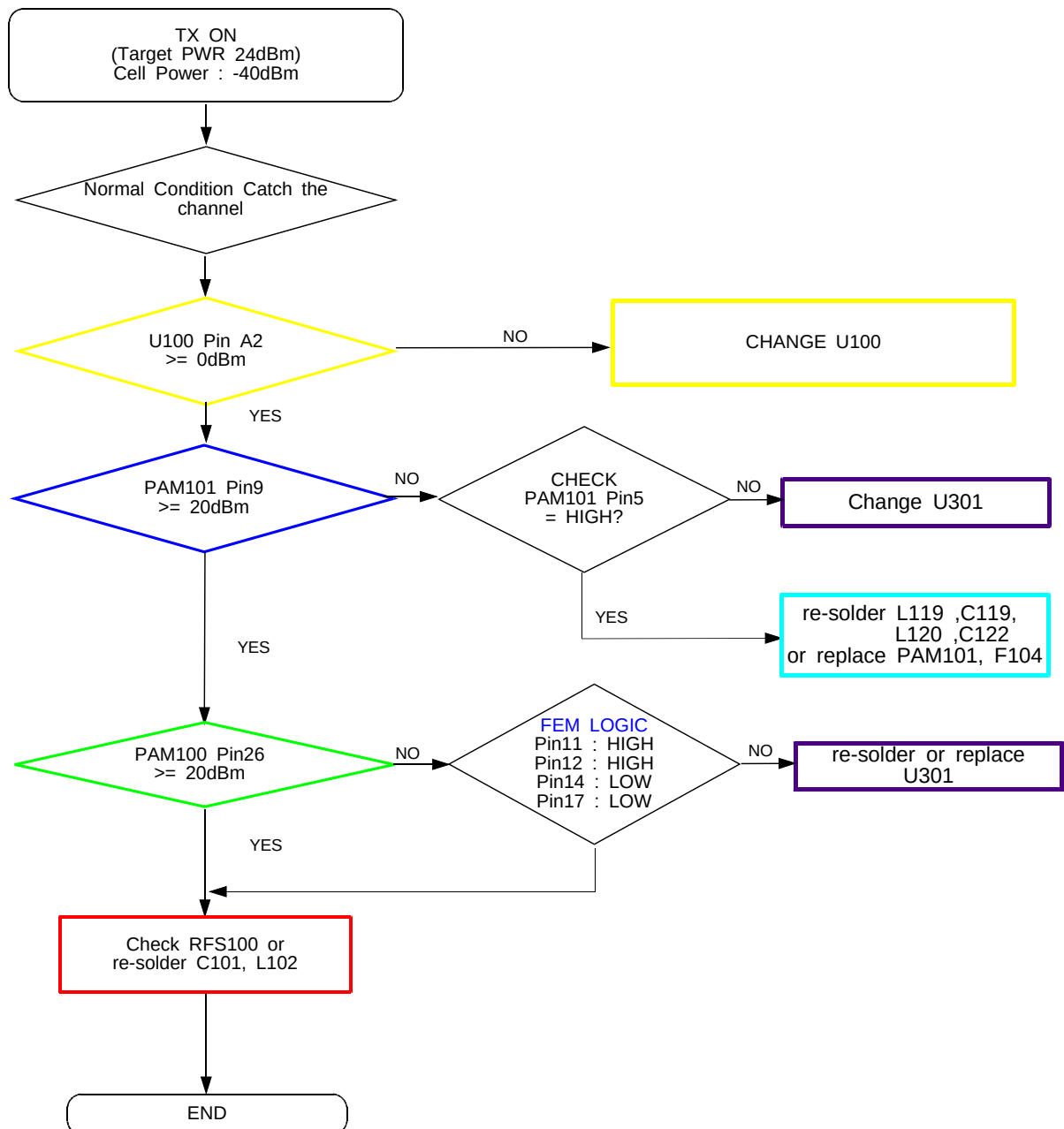
8-3-20. TDSCDMA B1,B2 Receiver

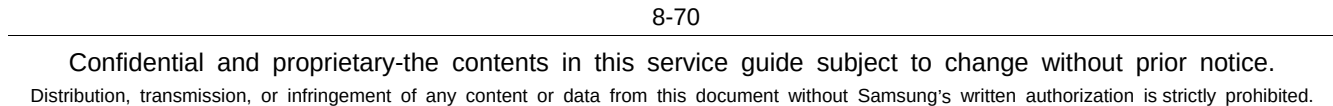


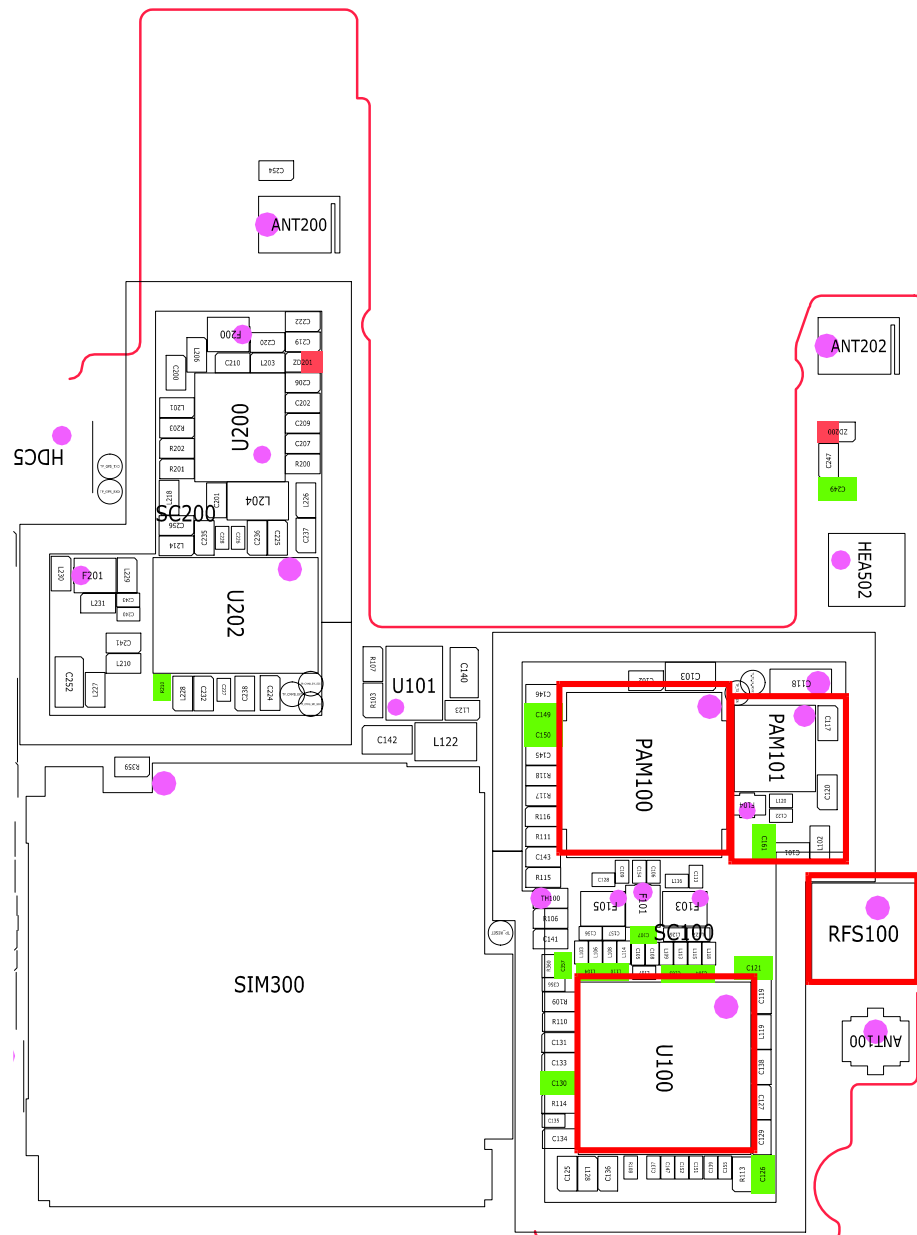




8-3-21. TDSCDMA B1,B2 Transmitter







8-4. Service Schematics (GT-I8250)

CP U301 (88CP920)

Bottom View (Unused pin)

● NC

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
A			●	●				●					●				●	●	●	●		
B	●	●	●	●				●					●	●		●	●	●	●			
C			●	●					●	●							●	●	●	●	●	●
D								●			●		●	●	●							
E																						
F											●				●							
G											●		●									
H												●	●									
J		●																				
K				●																		
L																						
M																						
N																						
P																						
R																						
T							●				●											
U											●			●								
V																						
W					●	●				●	●	●						●				
Y		●					●		●	●		●	●									
AA			●		●				●	●	●	●	●	●	●			●	●			
AB		●						●	●	●	●	●	●	●								

Top view (unused pin)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
A	A1_NC	A2_NC	MIPI_DAT A1_N	MIPI_DAT A1_P	GPIO_75	GPIO_73	GPIO_69	PA_MOD E	DIGRF_T X_P	DIGRF_T X_N	DIGRF_R X_N	DIGRF_R X_P	SYSCLOCK N	GPIO_65	GPIO_61	GPIO_60	TDS_DIO 6
B	VREFS	MIPI_DAT A0_P	MIPI_DAT A0_N	MIPI_CLK N	GPIO_76	GPIO_74	GPIO_70	RF_CON T_4	DIGRF_IR EF	DIGRF_A VSS	DIGRF_A VDD18	AVSS_PL LM	GSM_CD ATA	GSM_ST ROBE	GPIO_63	TDS_TXR EV	TDS_DIO 1
C	CAL_PAD	VCC_MAI N	MIPI_RC OMP	MIPI_CLK P	GPIO_78	GPIO_72	GPIO_71	GPIO_68	ANT_SW 4	ANT_SW 2	VCC_MAI N	AVDD18_ PLLM	SIN_0P4V	SYSCLOCK	GPIO_62	C16_NC	TDS_DIO 2
D	SDRAM_ DQ[0]	SDRAM_ DQ[2]	SDRAM_ DQM[0]	MIPI_AVS S	MIPI_AVDD D1P2V	VSS	VCC_IO_ GPIO2	VSS_IO_ ANT	ANT_SW 1	VCC_MAI N	TEST_PI N_PLLM	VSS	RXTXDAT A	GSM_CE N	GSM_CC LK	GPIO_66	D17_NC
E	VDDQ	SDRAM_ DQ[4]	SDRAM_ DQ[3]	SDRAM_ DQ[1]													
F	VSS	SDRAM_ DQ[7]	VDDQ	VSS		VCC18_I O_GPIO2	GPIO_79	VSS_IO_ GPIO2	VCC_IO_ ANT	VSS	ANT_SW 3	VCC_IO_ GSM	VSS_IO_ GSM	VCC_MAI N	TDS_DIO 0	VSS	GPIO_10 9
G	SDRAM_ DQS[0]	SDRAM_ DQ[8]	SDRAM_ DQM[12]	SDRAM_ DQ[5]		GPIO_77	GPIO_80	VCC_MAI N	GPIO_67	VCC18_I O_ANT	RXTXEN	GPIO_64	TDS_DIO 4	VSS	GPIO_10 1	GPIO_10 8	GPIO_10 7
H	SDRAM_ DQS[0]	SDRAM_ DQ[14]	VCC_MAI N	VDDQ		VSS	SDRAM_ DQ[9]	SDRAM_ DQM[1]	H9_NC	H10_NC	H11_NC	TDS_DIO 3	TDS_DIO 8	GPIO_98	GPIO_91	GPIO_94	GPIO_10 3
J	SDRAM_ DQS[1]	SDRAM_ CSN1	SDRAM_ DQ[15]	SDRAM_ DQ[13]		SDRAM_ DQ[6]	SDRAM_ DQ[10]	VSS							GPIO_85	GPIO_87	GPIO_10 5
K	SDRAM_ DQS[1]	VDDQ	SDRAM_ CKE0	SDRAM_ CKE1		SDRAM_ DQ[11]	VDDQ	VSS							VCC_IO_ GPIO1	VSS_IO_ GPIO1	GPIO_89
L	SDRAM_ CLK	SDRAM_ ADDR[0]	VSS	SDRAM_ ADDR[7]		VDDQ	SDRAM_ CSN0	SDRAM_ ADDR[3]							GPIO_40	GPIO_44	GPIO_83
M	SDRAM_ CLKN	SDRAM_ ADDR[2]	SDRAM_ ADDR[1]	VSS		VSS	SDRAM_ ADDR[5]	SDRAM_ ADDR[9]							GPIO_38	GPIO_32	GPIO_50
N	SDRAM_ DQS[2]	SDRAM_ ADDR[8]	SDRAM_ ADDR[6]	SDRAM_ DQM[2]		SDRAM_ DQ[17]	SDRAM_ DQ[23]	VSS							GPIO_00	GPIO_06	VSS_IO_ GPIO1
P	SDRAM_ DQS[2]	SDRAM_ ADDR[4]	SDRAM_ DQ[16]	SDRAM_ DQ[19]		SDRAM_ DQ[18]	SDRAM_ DQ[29]	VDDQ							VSS_IO_ PMIC2	VCC_MAI N	GPIO_10
R	SDRAM_ DQS[3]	SDRAM_ DQ[20]	SDRAM_ DQ[21]	SDRAM_ DQM[3]		SDRAM_ DQ[25]	MMC1_C MD	VSS	VCC_MAI N	ND_CLE SM_NOE	VCC_IO_ PMIC1	VSS_IO_ NAND	VSS_IO_ PMIC1	PRI_TCK	CLK_SRC SEL	VCC_IO_ GPIO1	GPIO_22
T	SDRAM_ DQS[3]	SDRAM_ DQ[24]	SDRAM_ DQ[22]	SDRAM_ DQ[27]		SDRAM_ DQ[31]	MMC1_W P	AVSS_US B	AVDD_U SB	VCC_IO_ NAND	ND_IO9	VCC18_I O_NAND	VCXO_E N	VCC_IO_ NAND	VSS	VCXO_R EQ	GPIO_24
U	SDRAM_ DQ[26]	VSS	VDDQ	VSS		MMC1_D AT1	VSS	USBVBU S	ND_IO1	ND_IO5	ND_IO13	VSS	SM_SCLK	SM_NBE0	PRI_TDI	VCC_MAI N	GPIO_02
V	VDDQ	SDRAM_ DQ[28]	SDRAM_ DQ[30]	VCC_IO_ USIM													
W	VCC_MAI N	USIM_UC LK	USIM_NU RST	MMC1_D AT6	MMC1_D AT4	MMC1_D AT2	MMC1_C LK	USB_P	USB_N	SM_NCS 0	ND_ALE SM_NWE	SM_ADV MUX	PWR_SD A	VCC_IO_ PMIC2	SLAVE_R ESET_OU T	VCXO_O UT	VSS_IO_ GPIO1
Y	VSS_IO_ USIM	MMC1_D AT5	USIM_UI O	VCC_IO_ MMC1	MMC1_D AT0	VCC_MAI N	USBID	ND_IO7	ND_IO12	ND_IO15	VSS_IO_ NAND	ND_RDY1	JTAG_SE L	EXT_32K IN	PMIC_INT	PRI_TDO	CLK_REQ
AA	AA1_NC	MMC1_D AT3	MMC1_D AT7	MMC1_C D	AVDD5_U SB	USB2_RE XT	ND_IO4	ND_IO6	ND_IO10	ND_NCS0 _SM_NC S2	ND_NWE	ND_NRE	SM_NBE1	SM_RDY	SLEEP_I ND	PRI_TMS	GPIO_12 4

BT WIFI CHIP U201(88W8787)

● NC

	1	2	3	4	5	6	7	8	9	10	11	12	13
A				●	●	●						●	●
B						●			●		●	●	
C		●										●	●
D	●							●		●		●	
E		●		●						●			
F				●						●			
G	●	●								●			
H												●	●
J	●											●	●
K					●		●			●			●
L													●
M					●		●					●	
N							●				●	●	